HSC Math Extension 2: Collection of Hard Problems

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Download PDF and resources:
`https://github.com/vuhung16au/math-olympiad-ml`

"Mathematics is the music of reason."

James Joseph Sylvester

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1 Overview

This collection presents a curated set of challenging problems from the HSC Mathematics Extension 2 curriculum. These problems are designed to test deep understanding, creative problem-solving skills, and the ability to synthesize multiple mathematical concepts.

1.1 What This Collection Is About

This collection focuses on **HSC Mathematics Extension 2 (hard problems)**. The problems span various topics including:

- Complex numbers and their geometric interpretations
- Integration techniques and applications
- Vector geometry in three dimensions
- Mechanics and particle motion
- Inequalities and optimization

Each problem is carefully selected to represent the level of difficulty and sophistication expected in the most challenging HSC Extension 2 examinations.

1.2 Target Audience

This collection is designed for:

- **Students** preparing for HSC Mathematics Extension 2 who want to challenge themselves with difficult problems
- **Tutors** seeking high-quality problems to use in their teaching
- **Educators** looking for challenging problems to incorporate into their curriculum

The problems are presented with hints to guide thinking, followed by detailed solutions and key takeaways to reinforce learning.

2 Problems

2.1 Problem 1: Two Particles in Resisting Medium

Problem 2.1

Two particles, A and B, each have mass 1 kg and are in a medium that exerts a resistance to motion equal to kv, where $k > 0$ and v is the velocity of any particle. Both particles maintain vertical trajectories.
The acceleration due to gravity is g m s^{-2}, where $g > 0$.
The two particles are simultaneously projected towards each other with the same speed, v_0 m s^{-1}, where $0 < v_0 < \frac{g}{k}$.
The particle A is initially d metres directly above particle B, where $d < \frac{2v_0}{k}$.
Find the time taken for the particles to meet.

Hint: Consider the equations of motion for each particle under the influence of gravity and resistance. Set up differential equations for the velocities and positions. The condition for the particles to meet will give you an equation involving time.

2.2 Problem 2: Complex Square with Equilateral Triangle

Problem 2.2

A square in the Argand plane has vertices

$$5+5i, \quad 5-5i, \quad -5-5i \quad \text{and} \quad -5+5i.$$

The complex numbers $z_A = 5 + i$, z_B and z_C lie on the square and form the vertices of an equilateral triangle.
Find the exact value of the complex number z_B.

Hint: Use the geometric properties of equilateral triangles. Consider rotations in the complex plane. The vertices of an equilateral triangle are related by rotations of 60° or 120° about the centroid.

2.3 Problem 3: Complex 7th Root of Unity

Problem 2.3

Let w be a complex number such that $1 + w + w^2 + \cdots + w^6 = 0$.

(i) Show that w is a 7th root of unity.

(ii) The complex number $\alpha = w + w^2 + w^4$ is a root of the equation $x^2 + bx + c = 0$, where b and c are real and α is not real. Find the other root of $x^2 + bx + c = 0$ in terms of positive powers of w.

(iii) Find the numerical value of c.

Hint: For part (i), use the formula for the sum of a geometric series. For part (ii), use the fact that for a quadratic with real coefficients, the other root is the complex conjugate. For part (iii), use properties of roots of unity and their relationships.

2.4 Problem 4: Complex Triangle Inequality

Problem 2.4

The complex number z satisfies $\left|z - \frac{4}{z}\right| = 2$.
Using the triangle inequality, or otherwise, show that $|z| \le \sqrt{5} + 1$.

Hint: Apply the triangle inequality to $|z - \frac{4}{z}|$. Consider both directions: $|z| - |\frac{4}{z}| \le |z - \frac{4}{z}|$ and $|z - \frac{4}{z}| \le |z| + |\frac{4}{z}|$. Use the given condition to derive bounds on $|z|$.

2.5 Problem 5: Integral with Inverse Sine

Problem 2.5

Using the substitution $x = \tan^2\theta$, evaluate

$$\int_0^1 \sin^{-1}\sqrt{\frac{x}{1+x}}\,dx.$$

Hint: After making the substitution, simplify the integrand. You may need to use trigonometric identities. Consider the relationship between $\sin^{-1}$ and the substitution variable.

2.6 Problem 6: Integral with Cotangent

Problem 2.6

Let $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{2n}\theta\,d\theta$ for integers $n \geq 0$.

(i) Show that $I_n = \frac{1}{2n-1} - I_{n-1}$ for $n > 0$, given that $\frac{d}{d\theta}\cot\theta = -\csc^2\theta$.

(ii) Hence, or otherwise, calculate I_2.

Hint: For part (i), use integration by parts. Write $\cot^{2n}\theta = \cot^{2n-2}\theta \cdot \cot^2\theta$ and use the identity $\cot^2\theta = \csc^2\theta - 1$. For part (ii), use the recurrence relation and find I_0 first.

2.7 Problem 7: 3D Vectors and Distance

Problem 2.7

Consider the point B with three-dimensional position vector $\mathbf{b}$ and the line l: $\mathbf{a} + \lambda\mathbf{d}$, where $\mathbf{a}$ and $\mathbf{d}$ are three-dimensional vectors, $|\mathbf{d}| = 1$ and λ is a parameter.
Let $f(\lambda)$ be the distance between a point on the line l and the point B.

(i) Find λ_0, the value of λ that minimises f, in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{d}$.

(ii) Let P be the point with position vector $\mathbf{a} + \lambda_0\mathbf{d}$. Show that PB is perpendicular to the direction of the line l.

(iii) Hence, or otherwise, find the shortest distance between the line l and the sphere of radius 1 unit, centred at the origin O, in terms of $\mathbf{d}$ and $\mathbf{a}$.

You may assume that if B is the point on the sphere closest to l, then OBP is a straight line.

Hint: For part (i), minimize $f^2(\lambda)$ by differentiating. For part (ii), use the fact that the minimum distance occurs when the vector from P to B is perpendicular to the direction vector. For part (iii), use the geometric interpretation and the given assumption.

2.8 Problem 8: Triangle Inequality and Rectangular Prism

Problem 2.8

It is given that for positive numbers $x_1, x_2, x_3, \ldots, x_n$ with arithmetic mean A,

$$\frac{x_1 \times x_2 \times x_3 \times \cdots \times x_n}{A^n} \leq 1. \quad \text{(Do NOT prove this.)}$$

Suppose a rectangular prism has dimensions a, b, c and surface area S.

(i) Show that $abc \leq \left(\frac{S}{6}\right)^{\frac{3}{2}}$.

(ii) Using part (i), show that when the rectangular prism with surface area S is a cube, it has maximum volume.

Hint: For part (i), relate the surface area to the dimensions and apply the given inequality. For part (ii), show that equality in the inequality from part (i) occurs when $a = b = c$, which corresponds to a cube.

2.9 Problem 9: Three Unit Vectors Optimization

Problem 2.9

Three unit vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, in 3 dimensions, are to be chosen so that $\mathbf{a} \perp \mathbf{b}$, $\mathbf{b} \perp \mathbf{c}$ and the angle θ between $\mathbf{a}$ and $\mathbf{a} + \mathbf{b} + \mathbf{c}$ is as small as possible.
What is the value of $\cos\theta$?

Hint: Use the dot product to express $\cos\theta$ in terms of the vectors. Consider the constraints and use Lagrange multipliers or geometric reasoning. The optimal configuration occurs when the vectors are arranged symmetrically.

2.10 Problem 10: Complex Numbers with Argument Condition

Problem 2.10

For the complex numbers z and w, it is known that $\arg\left(\frac{z}{w}\right) = -\frac{\pi}{2}$.
Find $\left|\frac{z-w}{z+w}\right|$.

Hint: The condition $\arg(z/w) = -\pi/2$ means z/w is purely imaginary and negative. Write $z = -ikw$ for some real $k > 0$, or use geometric interpretation. Then simplify the expression $|z - w|/|z + w|$.

2.11 Problem 11: Vectors and Complex Numbers

Problem 2.11

Consider the three vectors $\mathbf{a} = \vec{OA}$, $\mathbf{b} = \vec{OB}$ and $\mathbf{c} = \vec{OC}$, where O is the origin and the points A, B and C are all different from each other and the origin.
The point M is the point such that $\frac{1}{2}(\mathbf{a} + \mathbf{b}) = \vec{OM}$.

(i) Show that M lies on the line passing through A and B.

(ii) The point G is the point such that $\frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \vec{OG}$. Show that G lies on the line passing through M and C, and lies between M and C.

(iii) The complex numbers x, w and z are all different and all have modulus 1. Using part (ii), or otherwise, show that $\frac{1}{3}(x + w + z)$ is never a cube root of xwz.

Hint: For part (i), express M as a linear combination of A and B. For part (ii), show that G can be written as a weighted combination of M and C. For part (iii), use the geometric interpretation: if $\frac{1}{3}(x + w + z)$ were a cube root of xwz, what would that imply about the positions on the unit circle?

2.12 Problem 12: Bar Magnet and Falling Object

Problem 2.12

A bar magnet is held vertically. An object that is repelled by the magnet is to be dropped from directly above the magnet and will maintain a vertical trajectory.
Let x be the distance of the object above the magnet.
The object is subject to acceleration due to gravity, g, and an acceleration due to the magnet so that the total acceleration of the object is given by

$$a = \frac{27g}{x^3} - g.$$

The object is released from rest at $x = 6$.

(i) Show that $v^2 = g\left(\frac{51}{4} - 2x - \frac{27}{x^2}\right)$.

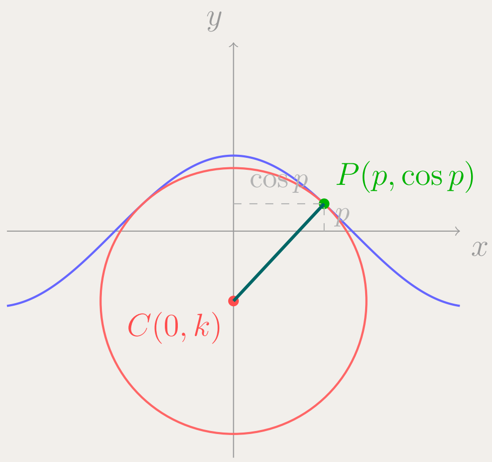
(ii) Find where the object next comes to rest, giving your answer correct to 1 decimal place.

Hint: For part (i), use $a = v\frac{dv}{dx}$ and integrate. For part (ii), set $v = 0$ and solve the resulting equation. You may need to use numerical methods or factor the polynomial.

2.13 Problem 13: Circle and Cosine Function

Problem 2.13

Consider the function $y = \cos(kx)$ where $k > 0$. The value of k has been chosen so that a circle can be drawn, centred at the origin, which has exactly two points of intersection with the graph of the function and so that the circle is never above the graph of the function.
The point $P(a, b)$ is the point of intersection in the first quadrant, so $a > 0$ and $b > 0$, as shown in the diagram below.



The vector joining the origin to the point $P(a, b)$ is perpendicular to the tangent to the graph of the function at that point. (Do NOT prove this.)
Show that $k > 1$.

Hint: At point $P(a, b)$, the radius vector is perpendicular to the tangent. The slope of the radius is b/a, and the slope of the tangent is $-k\sin(ka)$. Use the perpendicularity condition and the fact that P lies on both the circle and the curve.

2.14 Problem 14: Cube Roots of Unity and Trigonometric Products

Problem 2.14

The number $w = e^{\frac{2\pi i}{3}}$ is a complex cube root of unity. The number γ is a cube root of w.

(i) Show that $\gamma + \overline{\gamma}$ is a real root of $z^3 - 3z + 1 = 0$.

(ii) By using part (i) to find the exact value of

$$\cos\frac{2\pi}{9}\cos\frac{4\pi}{9}\cos\frac{8\pi}{9},$$

deduce the value(s) of $\cos\frac{2^n\pi}{9}\cos\frac{2^{n+1}\pi}{9}\cos\frac{2^{n+2}\pi}{9}$ for all integers $n \geq 1$. Justify your answer.

Hint: For part (i), if $\gamma^3 = \omega$, then γ is a 9th root of unity. Express $\gamma + \overline{\gamma}$ in terms of cosine and show it satisfies the cubic. For part (ii), use trigonometric identities and the fact that 2^n mod 9 cycles through certain values.

2.15 Problem 15: Integer Equation with Large Exponents

Problem 2.15

Explain why there is no integer n such that $(n+1)^{41} - 79n^{40} = 2$.

Hint: Consider the equation modulo a small prime number. Try working modulo 2, 3, or other small primes. Also consider the binomial expansion of $(n+1)^{41}$ and look for divisibility properties.

2.16 Problem 16: Projectile with Quadratic Resistance

Problem 2.16

A projectile of mass M kg is launched vertically upwards from the origin with an initial speed v_0 m s^{-1}. The acceleration due to gravity is g m s^{-2}.
The projectile experiences a resistive force of magnitude kMv^2 newtons, where k is a positive constant and v is the speed of the projectile at time t seconds.

(i) The maximum height reached by the particle is H metres. Show that

$$H = \frac{1}{2k} \ln\left(\frac{kv_0^2 + g}{g}\right).$$

(ii) When the projectile lands on the ground, its speed is v_1 m s^{-1}, where v_1 is less than the magnitude of the terminal velocity. Show that $g(v_0^2 - v_1^2) = kv_0^2 v_1^2$.

Hint: For part (i), use $a = v\frac{dv}{dx}$ and integrate. The resistive force opposes motion, so the acceleration is $-g - kv^2$ on the way up. For part (ii), consider the energy or use the same technique for the downward motion.

2.17 Problem 17: Integration with Recurrence Relations

Problem 2.17

(i) Let
$$J_n = \int_0^{\frac{\pi}{2}} \sin^n \theta \, d\theta$$
where $n \geq 0$ is an integer. Show that $J_n = \frac{n-1}{n} J_{n-2}$ for all integers $n \geq 2$.

(ii) Let
$$I_n = \int_0^1 x^n (1-x)^n dx$$
where n is a positive integer. By using the substitution $x = \sin^2 \theta$, or otherwise, show that
$$I_n = \frac{1}{2^{2n}} \int_0^{\frac{\pi}{2}} \sin^{2n+1} \theta \, d\theta.$$

(iii) Hence, or otherwise, show that $I_n = \frac{n}{4n+2} I_{n-1}$, for all integers $n \geq 1$.

Hint: For part (i), use integration by parts with $u = \sin^{n-1}\theta$ and $dv = \sin\theta \, d\theta$. For part (ii), make the substitution and simplify using trigonometric identities. For part (iii), combine the results from parts (i) and (ii).

2.18 Problem 18: Curve on a Sphere

Problem 2.18

A curve C spirals 3 times around the sphere centred at the origin and with radius 3, as shown below.

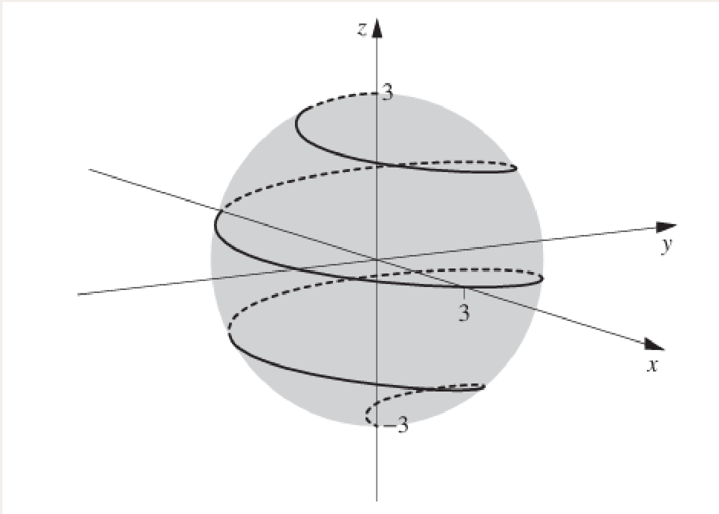


Figure 1: Curve C spiraling around the sphere

A particle is initially at the point $(0, 0, -3)$ and moves along the curve C on the surface of the sphere, ending at the point $(0, 0, 3)$.
By using the diagram below, which shows the graphs of the functions $f(x) = \cos(\pi x)$ and $g(x) = \sqrt{9 - x^2}$, and considering the graph $y = f(x)g(x)$, give a possible set of parametric equations that describe the curve C.

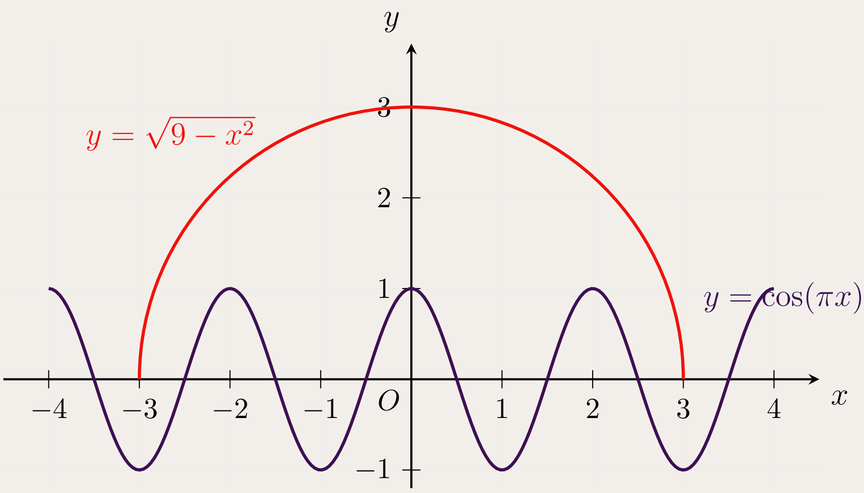


Figure 2: Graphs of $f(x) = \cos(\pi x)$ and $g(x) = \sqrt{9 - x^2}$

Hint: The curve is on a sphere of radius 3, so $x^2 + y^2 + z^2 = 9$. The function $g(x) = \sqrt{9 - x^2}$ suggests a relationship with the sphere. Use $f(x) = \cos(\pi x)$ to create the spiraling effect. Consider parameterizing with t such that z goes from -3 to 3 as the parameter increases.

2.19 Problem 19: Complex Numbers and Equilateral Triangles

Problem 2.19

Let w be the complex number

$$w = e^{\frac{2i\pi}{3}}.$$

(i) Show that $1 + w + w^2 = 0$.

(ii) Three complex numbers a, b and c are represented in the complex plane by points A, B and C respectively. Show that if triangle ABC is anticlockwise and equilateral, then $a + bw + cw^2 = 0$.

(iii) It can be shown that if triangle ABC is clockwise and equilateral, then $a + bw^2 + cw = 0$. (Do NOT prove this.) Show that if ABC is an equilateral triangle, then

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

Hint: For part (i), use the geometric series formula or note that w is a cube root of unity. For part (ii), use the fact that rotating an equilateral triangle by 120° maps vertices to each other. For part (iii), use both the anticlockwise and clockwise conditions, and manipulate the equations.

2.20 Problem 20: Inequalities with Exponentials and Factorials

Problem 2.20

(i) Prove that $x > \ln x$ for $x > 0$.

(ii) Using part (i), or otherwise, prove that for all positive integers n,

$$e^{n^2+n} > (n!)^2.$$

Hint: For part (i), consider the function $f(x) = x - \ln x$ and find its minimum. For part (ii), apply the inequality from part (i) to $x = 1, 2, \ldots, n$ and combine the results, using properties of exponentials and factorials.

2.21 Problem 21: Complex Numbers and Region Sketching

Problem 2.21

The complex numbers w and z both have modulus 1, and

$$\frac{\pi}{2} < \operatorname{Arg}\left(\frac{z}{w}\right) < \pi,$$

where Arg denotes the principal argument.
For real numbers x and y, consider the complex number

$$\frac{xz + yw}{z}.$$

On an xy-plane, clearly sketch the region that contains all points (x, y) for which

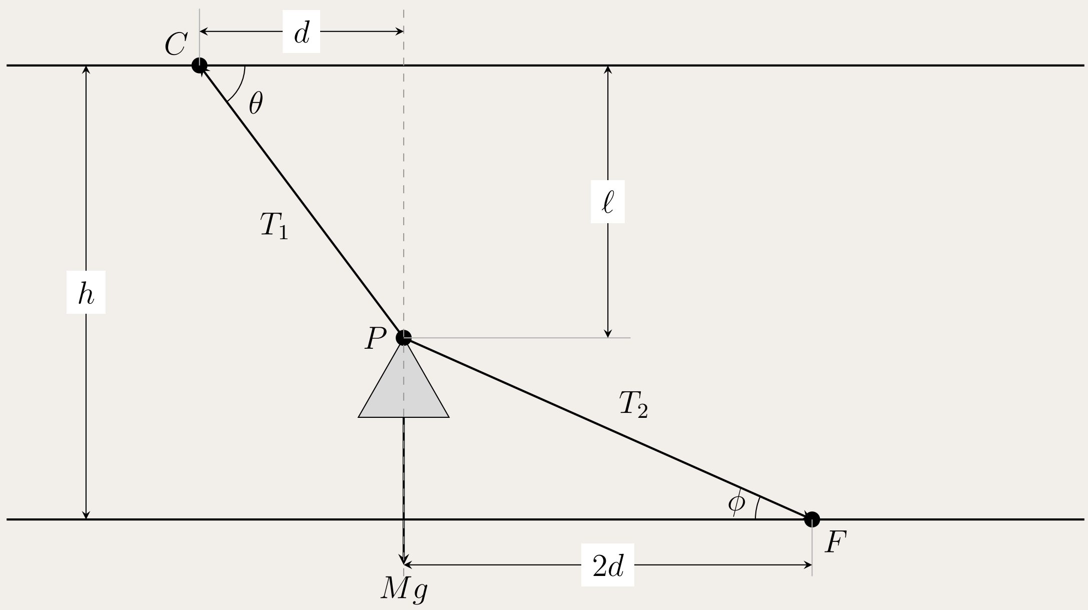
$$\frac{\pi}{2} < \operatorname{Arg}\left(\frac{xz + yw}{z}\right) < \pi.$$

Hint: Simplify $\frac{xz+yw}{z} = x + y\frac{w}{z}$. Since $|\frac{w}{z}| = 1$ and $\operatorname{Arg}(\frac{z}{w}) \in (\pi/2, \pi)$, determine $\operatorname{Arg}(\frac{w}{z})$. Then find conditions on x and y such that the argument of $x + y\frac{w}{z}$ lies in $(\pi/2, \pi)$.

2.22 Problem 22: Mechanics with Ropes and Forces

Problem 2.22

A machine P of mass M kg is being lifted using two ropes. Rope T_1 is attached to the ceiling C and rope T_2 is attached to the floor F. The ropes make angles θ and ϕ with the horizontal respectively, as shown in the diagram below.



(i) By considering horizontal and vertical components of the forces at P, show that

$$\tan\theta = \tan\phi + \frac{Mg}{T_2\cos\phi}.$$

(ii) Hence, or otherwise, show that the point P cannot be lifted to a position $\frac{2h}{3}$ metres above the floor.

Hint: For part (i), resolve forces horizontally and vertically at point P. For part (ii), use the relationship from part (i) and consider the geometry constraints involving the heights and angles.

2.23 Problem 23: Simple Harmonic Motion

Problem 2.23

A piston moves with simple harmonic motion between a maximum height of 0.17 m and a minimum height of 0.05 m.
The mass of the piston is 0.8 kg. The piston completes 40 cycles per second.
What is the resultant force on the piston, in newtons, that produces the maximum acceleration of the piston? Give your answer correct to the nearest newton.

Hint: For SHM, the maximum acceleration occurs at the extremes. Use $a_{\max} = \omega^2 A$ where $\omega = 2\pi f$ and A is the amplitude. Then use $F = ma$.

2.24 Problem 24: Projectile with Linear Resistance

Problem 2.24

A projectile of mass M kg is launched vertically upwards from a horizontal plane with initial speed v_0 m s^{-1} which is less than 100 m s^{-1}.
The projectile experiences a resistive force which has magnitude $0.1Mv$ newtons, where v m s^{-1} is the speed of the projectile.
The acceleration due to gravity is 10 m s^{-2}.
The projectile lands on the horizontal plane 7 seconds after launch.
Find the value of v_0, correct to 1 decimal place.

Hint: Set up the differential equation for motion with linear resistance: $\frac{dv}{dt} = -g - 0.1v$. Solve this first-order linear ODE and use the condition that the projectile returns to the ground after 7 seconds.

2.25 Problem 25: Complex Numbers with Three Conditions

Problem 2.25

Find all the complex numbers z_1, z_2, z_3 that satisfy the following three conditions simultaneously:

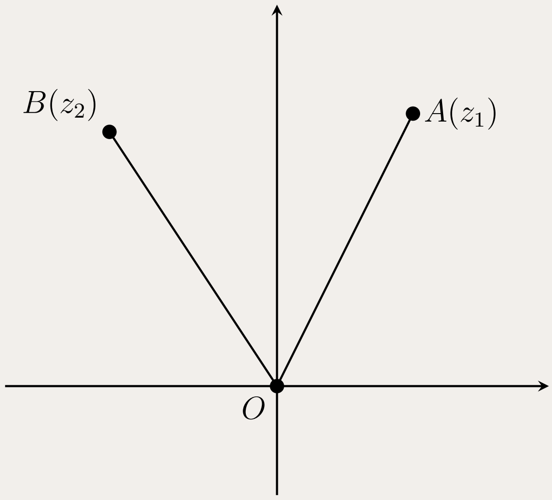
$$\begin{cases} |z_1| = |z_2| = |z_3| \\ z_1 + z_2 + z_3 = 1 \\ z_1 z_2 z_3 = 1 \end{cases}$$

Hint: Since all have the same modulus, write $z_k = r(\cos\theta_k + i\sin\theta_k)$. Use the sum and product conditions. Consider the geometric interpretation: three complex numbers on a circle that sum to 1 and multiply to 1.

2.26 Problem 26: Complex Numbers and Equilateral Triangle

Problem 2.26

Let z_1 be a complex number and let $z_2 = \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) z_1$. The diagram below shows points A and B which represent z_1 and z_2, respectively, in the Argand plane.



(i) Explain why triangle OAB is an equilateral triangle.

(ii) Prove that ${z_1}^2 + {z_2}^2 = z_1 z_2$.

Hint: For part (i), note that multiplying by $\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$ rotates by $60°$. For part (ii), substitute $z_2 = \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) z_1$ and use $\cos\frac{\pi}{3} = \frac{1}{2}$, $\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$.

2.27 Problem 27: Particle Falling with Resistance

Problem 2.27

A particle starts from rest and falls through a resisting medium so that its acceleration, in m/s^2, is modelled by

$$a = 10(1 - (kv)^2),$$

where v is the velocity of the particle in m/s and $k = 0.01$.
Find the velocity of the particle after 5 seconds.

Hint: Use $a = \frac{dv}{dt}$ and separate variables. The equation is $\frac{dv}{dt} = 10(1 - 0.0001v^2)$. This is a separable differential equation. Find the terminal velocity first.

2.28 Problem 28: Induction Proof for Series

Problem 2.28

Prove by mathematical induction that, for $n \geq 2$,

$$\frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < \frac{n-1}{n}.$$

Hint: For the base case, check $n = 2$. For the inductive step, assume the inequality holds for $n = k$ and show it holds for $n = k + 1$ by adding $\frac{1}{(k+1)^2}$ to both sides and manipulating the inequality.

2.29 Problem 29: Irrationality of Logarithm

Problem 2.29

Prove that for any integer $n > 1$, $\log_n(n+1)$ is irrational.

Hint: Use proof by contradiction. Assume $\log_n(n+1) = \frac{p}{q}$ for integers p, q. Then $n^{p/q} = n+1$, so $n^p = (n+1)^q$. Use the fact that n and $n+1$ are coprime.

2.30 Problem 30: Proposition Logic

Problem 2.30

In the set of integers, let P be the proposition: 'If $k + 1$ is divisible by 3, then $k^3 + 1$ is divisible by 3'.

(i) Prove that the proposition P is true.

(ii) Write down the contrapositive of the proposition P.

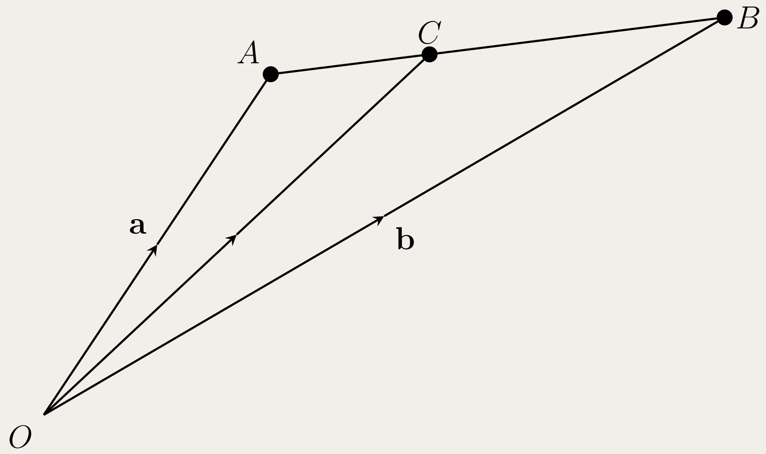
(iii) Write down the converse of the proposition P and state, with reasons, whether this converse is true or false.

Hint: For part (i), if $k + 1 = 3m$, then $k = 3m - 1$. Expand $k^3 + 1$ and show it's divisible by 3. For part (ii), the contrapositive is 'If $k^3 + 1$ is not divisible by 3, then $k + 1$ is not divisible by 3'. For part (iii), test the converse with counterexamples.

2.31 Problem 31: Vectors and Ratios

Problem 2.31

The point C divides the interval AB so that $\frac{CB}{AC} = \frac{m}{n}$. The position vectors of A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively, as shown in the diagram below.



(i) Show that $\vec{AC} = \frac{n}{m+n}(\mathbf{b} - \mathbf{a})$.

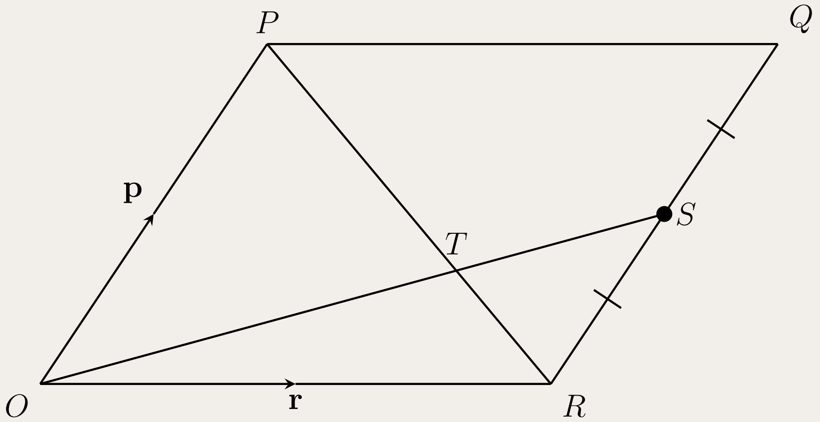
(ii) Prove that $\vec{OC} = \frac{m}{m+n}\mathbf{a} + \frac{n}{m+n}\mathbf{b}$.

Hint: For part (i), use the ratio $\frac{CB}{AC} = \frac{m}{n}$ and the fact that $AC + CB = AB$. For part (ii), use $\vec{OC} = \vec{OA} + \vec{AC}$.

2.32 Problem 32: Parallelogram Geometry

Problem 2.32

Let $OPQR$ be a parallelogram with $\vec{OP} = \mathbf{p}$ and $\vec{OR} = \mathbf{r}$. The point S is the midpoint of QR and T is the intersection of PR and OS, as shown in the diagram below.



(i) Show that $\vec{OT} = \frac{2}{3}\mathbf{r} + \frac{1}{3}\mathbf{p}$.

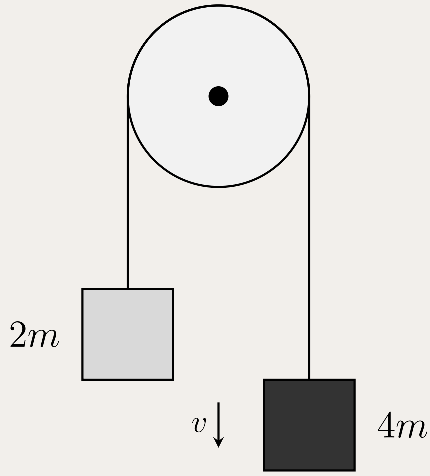
(ii) Using part (i), or otherwise, prove that T is the point that divides the interval PR in the ratio 2:1.

Hint: For part (i), express T as a point on both lines PR and OS using parameters, then equate. For part (ii), show that $\vec{PT} = 2\vec{TR}$ or use the section formula.

2.33 Problem 33: Two Masses with Pulley

Problem 2.33

Two masses, $2m$ kg and $4m$ kg, are attached by a light string. The string is placed over a smooth pulley as shown below. The two masses are at rest before being released and v is the velocity of the larger mass at time t seconds after they are released.



The force due to air resistance on each mass has magnitude kv, where k is a positive constant.

(i) Show that

$$\frac{dv}{dt} = \frac{gm - kv}{3m}.$$

(ii) Given that $v < \frac{gm}{k}$, show that when $t = \frac{3m}{k}\ln 2$, the velocity of the larger mass is $\frac{gm}{2k}$.

Hint: For part (i), apply Newton's second law to each mass, considering tensions and resistance. For part (ii), solve the differential equation and substitute the given time.

2.34 Problem 34: Integration with Recurrence and Factorial Inequality

Problem 2.34

Let $I_n = \int_0^{\frac{\pi}{2}} \sin^{2n+1}(2\theta)\, d\theta$, $n = 0, 1, \ldots$.

(i) Prove that $I_n = \frac{2n}{2n+1} I_{n-1}$, $n \geq 1$.

(ii) Deduce that $I_n = \frac{2^{2n}(n!)^2}{(2n+1)!}$.

(iii) Let $J_n = \int_0^1 x^n(1-x)^n\, dx$, $n = 0, 1, 2, \ldots$. Using the result of part (ii), or otherwise, show that

$$J_n = \frac{(n!)^2}{(2n+1)!}.$$

(iv) Prove that $(2^n n!)^2 \leq (2n+1)!$.

Hint: For part (i), use integration by parts. For part (ii), use the recurrence and find I_0. For part (iii), use substitution $x = \sin^2\theta$ to relate J_n to I_n. For part (iv), use the fact that $J_n \leq 1$ (since the integrand is bounded by 1 on [0, 1]).

2.35 Problem 35: Basel problem, solved by Leonhard Euler in 1734

Problem 2.35

For integer $n \geq 0$. Let

$$A_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x\, dx \quad \text{and} \quad B_n = \int_0^{\frac{\pi}{2}} x^2 \cos^{2n} x\, dx$$

(a) Show that

$$nA_n = \frac{2n-1}{2} A_{n-1} \quad \text{for } n \geq 1.$$

(b) Using integration by parts on A_n, or otherwise, show that:

$$A_n = 2n \int_0^{\frac{\pi}{2}} x \sin x \cos^{2n-1} x\, dx \quad \text{for } n \geq 1.$$

(c) Use integration by parts on the integral in part (b) to show that:

$$\frac{A_n}{n^2} = \frac{(2n-1)}{n} B_{n-1} - 2B_n \quad \text{for } n \geq 1.$$

(d) Use parts (a) and (c) to show that

$$\frac{1}{n^2} = 2\left(\frac{B_{n-1}}{A_{n-1}} - \frac{B_n}{A_n}\right) \quad \text{for } n \geq 1.$$

(e) Show that

$$\sum_{k=1}^{n} \frac{1}{k^2} = \frac{\pi^2}{6} - 2\frac{B_n}{A_n}.$$

(f) Use the fact that $\sin x \geq \frac{2}{\pi} x$ for $0 \leq x \leq \frac{\pi}{2}$ to show that

$$B_n \leq \int_0^{\frac{\pi}{2}} x^2 \left(1 - \frac{4x^2}{\pi^2}\right)^n dx.$$

(g) Show that

$$\int_0^{\frac{\pi}{2}} x^2 \left(1 - \frac{4x^2}{\pi^2}\right)^n dx = \frac{\pi^2}{8(n+1)} \int_0^{\frac{\pi}{2}} \left(1 - \frac{4x^2}{\pi^2}\right)^{n+1} dx.$$

(h) From parts (f) and (g) it follows that

$$B_n \leq \frac{\pi^2}{8(n+1)} \int_0^{\frac{\pi}{2}} \left(1 - \frac{4x^2}{\pi^2}\right)^{n+1} dx.$$

Use the substitution $x = \frac{\pi}{2} \sin t$ in this inequality to show that

$$B_n \leq \frac{\pi^3}{16(n+1)} \int_0^{\frac{\pi}{2}} \cos^{2n+3} t\, dt \leq \frac{\pi^3}{16(n+1)} A_n.$$

(i) Use part (e) to deduce that

$$\frac{\pi^2}{6} - \frac{\pi^3}{8(n+1)} \leq \sum_{k=1}^{n} \frac{1}{k^2} < \frac{\pi^2}{6}.$$

(j) What is $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k^2}$?

3 Solutions

3.1 Solution to Problem 1: Two Particles in Resisting Medium

Solution 3.1

Let $y_A(t)$ and $y_B(t)$ be the positions of particles A and B respectively, with $y_A(0) = d$ and $y_B(0) = 0$. The equation of motion for each particle is:

$$\frac{dv}{dt} = -g - kv$$

Solving this first-order linear ODE with $v(0) = -v_0$ (downward for A) and $v(0) = v_0$ (upward for B):

$$v(t) = -\frac{g}{k} + \left(v_0 + \frac{g}{k}\right) e^{-kt}$$

Integrating to find position:

$$y(t) = y(0) - \frac{g}{k}t + \frac{1}{k}\left(v_0 + \frac{g}{k}\right)(1 - e^{-kt})$$

For particle A: $y_A(t) = d - \frac{g}{k}t + \frac{1}{k}\left(v_0 + \frac{g}{k}\right)(1 - e^{-kt})$
For particle B: $y_B(t) = \frac{g}{k}t - \frac{1}{k}\left(v_0 + \frac{g}{k}\right)(1 - e^{-kt})$
Setting $y_A(t) = y_B(t)$ and solving:

$$d = \frac{2}{k}\left(v_0 + \frac{g}{k}\right)(1 - e^{-kt})$$

Therefore, $t = -\frac{1}{k}\ln\left(1 - \frac{kd}{2(v_0+g/k)}\right)$.

Takeaways 3.1

- Motion with linear resistance: $dv/dt = -g - kv$ has solution $v = -g/k + (v_0 + g/k)e^{-kt}$
- Relative motion: Set positions equal to find meeting time
- Exponential decay in velocity due to resistance

3.2 Solution to Problem 2: Complex Square with Equilateral Triangle

Solution 3.2

The square has vertices $5+5i$, $5-5i$, $-5-5i$, and $-5+5i$, so its sides lie on the lines $\mathrm{Re}(z) = \pm 5$ and $\mathrm{Im}(z) = \pm 5$.
We are given $z_A = 5 + i$, which lies on the right edge $\mathrm{Re}(z) = 5$. The three points z_A, z_B, and z_C form an equilateral triangle, so they are related by 120° rotations about the centroid

$$z_G = \frac{z_A + z_B + z_C}{3}.$$

Multiplying by $e^{2\pi i/3}$ or $e^{-2\pi i/3}$ rotates a vertex to the next one. Since z_B and z_C must also lie on the square's perimeter, we test which edges they can occupy.
Using the rotation property $z_B - z_G = e^{2\pi i/3}(z_A - z_G)$ together with the constraint that z_B lies on one of the four sides of the square, we find

$$z_B = 5 - 5\sqrt{3} + (5\sqrt{3} - 4)i$$

(one valid orientation; the symmetric configuration gives the other vertex on the square).

Takeaways 3.2

- Equilateral triangles: vertices are $120°$ rotations of each other about the centroid
- Complex rotations: multiply by $e^{\pm 2\pi i/3}$ to rotate through $\pm 120°$
- Constraints: each vertex must lie on one of the four sides of the square

3.3 Solution to Problem 3: Complex 7th Root of Unity

Solution 3.3

(i) If $w = 1$, then $1 + w + \cdots + w^6 = 7 \neq 0$. So $w \neq 1$. Using the geometric series formula:

$$1 + w + w^2 + \cdots + w^6 = \frac{1 - w^7}{1 - w} = 0$$

Since $w \neq 1$, we have $1 - w^7 = 0$, so $w^7 = 1$. Therefore w is a 7th root of unity.
(ii) Since the quadratic has real coefficients, the other root is $\overline{\alpha} = \overline{w + w^2 + w^4} = \overline{w} + \overline{w^2} + \overline{w^4}$.
For a 7th root of unity $w = e^{2\pi ik/7}$ where $k \in \{1, 2, 3, 4, 5, 6\}$, we have $\overline{w} = w^{-1} = w^6$. Similarly, $\overline{w^2} = w^5$ and $\overline{w^4} = w^3$.
Therefore, $\overline{\alpha} = w^6 + w^5 + w^3$.
(iii) For $w = e^{2\pi i/7}$, we have:

$$c = \alpha \cdot \overline{\alpha} = (w + w^2 + w^4)(w^6 + w^5 + w^3)$$

Expanding and using $w^7 = 1$:

$$c = w^7 + w^6 + w^5 + w^8 + w^7 + w^6 + w^{10} + w^9 + w^7 = 3 + (w + w^2 + w^3 + w^4 + w^5 + w^6)$$

Since $1 + w + w^2 + \cdots + w^6 = 0$, we get $w + w^2 + \cdots + w^6 = -1$, so $c = 3 - 1 = 2$.

Takeaways 3.3

- 7th roots of unity: $w^7 = 1$ with $w \neq 1$ implies $1 + w + \cdots + w^6 = 0$
- Complex conjugate: For $w = e^{2\pi ik/7}$, we have $\overline{w} = w^{-1} = w^6$
- Product of roots: For real-coefficient quadratics, $c = \alpha\overline{\alpha} = |\alpha|^2$

3.4 Solution to Problem 4: Complex Triangle Inequality

Solution 3.4

Let $r = |z| > 0$. We are given $\left|z - \frac{4}{z}\right| = 2$. Apply the triangle inequality in both directions.

Lower bound. For any complex numbers,

$$\left|z - \frac{4}{z}\right| \geq \left||z| - \left|\frac{4}{z}\right|\right| = |r - \tfrac{4}{r}|.$$

So $2 \geq r - \frac{4}{r}$ (since $r > 0$ and typically $r \geq \frac{4}{r}$ for large $|z|$). Multiplying by $r > 0$,

$$2r \geq r^2 - 4 \quad \Longrightarrow \quad r^2 - 2r - 4 \leq 0.$$

The positive root of $r^2 - 2r - 4 = 0$ is $r = 1 + \sqrt{5}$, so

$$|z| \leq 1 + \sqrt{5}.$$

Upper bound. Also,

$$\left|z - \frac{4}{z}\right| \leq |z| + \left|\frac{4}{z}\right| = r + \frac{4}{r}.$$

So $2 \leq r + \frac{4}{r}$, giving $r^2 - 2r + 4 \geq 0$. This quadratic has discriminant $(-2)^2 - 4(4) = -12 < 0$, so it is always true and gives no further restriction on $|z|$.
Combining the useful bound,

$$|z| \leq 1 + \sqrt{5} = \sqrt{5} + 1.$$

Takeaways 3.4

- Triangle inequality: $|a - b| \geq \big||a| - |b|\big|$ and $|a - b| \leq |a| + |b|$
- Substitute $|z| = r$ to obtain quadratic inequalities in r
- Only the lower-bound direction produces a finite upper limit on $|z|$

3.5 Solution to Problem 5: Integral with Inverse Sine

Solution 3.5

Let $x = \tan^2\theta$, so $dx = 2\tan\theta\sec^2\theta\,d\theta = 2\tan\theta(1 + \tan^2\theta)\,d\theta$.
When $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \pi/4$.
The integrand becomes:

$$\sin^{-1}\sqrt{\frac{x}{1+x}} = \sin^{-1}\sqrt{\frac{\tan^2\theta}{1+\tan^2\theta}} = \sin^{-1}\sqrt{\frac{\tan^2\theta}{\sec^2\theta}} = \sin^{-1}|\sin\theta| = \sin^{-1}(\sin\theta) = \theta$$

for $\theta \in [0, \pi/4]$.
Therefore:

$$\int_0^1 \sin^{-1}\sqrt{\frac{x}{1+x}}\,dx = \int_0^{\pi/4} \theta \cdot 2\tan\theta\sec^2\theta\,d\theta$$

Using integration by parts with $u = \theta$, $dv = 2\tan\theta\sec^2\theta\,d\theta$:

$$= \left[\theta\tan^2\theta\right]_0^{\pi/4} - \int_0^{\pi/4} \tan^2\theta\,d\theta = \frac{\pi}{4} - \int_0^{\pi/4}(\sec^2\theta - 1)\,d\theta$$

$$= \frac{\pi}{4} - [\tan\theta - \theta]_0^{\pi/4} = \frac{\pi}{4} - (1 - \frac{\pi}{4}) = \frac{\pi}{2} - 1$$

Takeaways 3.5

- Substitution $x = \tan^2\theta$ simplifies $\sqrt{x/(1+x)}$ to $|\sin\theta|$
- For $\theta \in [0, \pi/4]$, we have $\sin^{-1}(\sin\theta) = \theta$
- Integration by parts: differentiate the polynomial part, integrate the trigonometric part

3.6 Solution to Problem 6: Integral with Cotangent

Solution 3.6

(i) Write $I_n = \int_{\pi/4}^{\pi/2} \cot^{2n}\theta\, d\theta = \int_{\pi/4}^{\pi/2} \cot^{2n-2}\theta \cdot \cot^2\theta\, d\theta$.
Using $\cot^2\theta = \csc^2\theta - 1$:

$$I_n = \int_{\pi/4}^{\pi/2} \cot^{2n-2}\theta(\csc^2\theta - 1)\, d\theta = \int_{\pi/4}^{\pi/2} \cot^{2n-2}\theta \csc^2\theta\, d\theta - I_{n-1}$$

For the first integral, let $u = \cot\theta$, so $du = -\csc^2\theta\, d\theta$:

$$\int_{\pi/4}^{\pi/2} \cot^{2n-2}\theta \csc^2\theta\, d\theta = -\int_1^0 u^{2n-2}\, du = \int_0^1 u^{2n-2}\, du = \frac{1}{2n-1}$$

Therefore: $I_n = \frac{1}{2n-1} - I_{n-1}$.
(ii) First, $I_0 = \int_{\pi/4}^{\pi/2} d\theta = \frac{\pi}{4}$.
Using the recurrence: $I_1 = \frac{1}{1} - I_0 = 1 - \frac{\pi}{4}$.
Then: $I_2 = \frac{1}{3} - I_1 = \frac{1}{3} - (1 - \frac{\pi}{4}) = \frac{\pi}{4} - \frac{2}{3}$.

Takeaways 3.6

- Recurrence: Use $\cot^2\theta = \csc^2\theta - 1$ to relate I_n and I_{n-1}
- Substitution: $u = \cot\theta$ converts $\cot^{2n-2}\theta \csc^2\theta\, d\theta$ to $u^{2n-2}\, du$
- Base case: $I_0 = \pi/4$ (integral of 1 over the interval)

3.7 Solution to Problem 7: 3D Vectors and Distance

Solution 3.7

(i) The distance squared is $f^2(\lambda) = |(\mathbf{a} + \lambda\mathbf{d}) - \mathbf{b}|^2 = |\mathbf{a} - \mathbf{b} + \lambda\mathbf{d}|^2$.
Expanding: $f^2(\lambda) = |\mathbf{a} - \mathbf{b}|^2 + 2\lambda(\mathbf{a} - \mathbf{b}) \cdot \mathbf{d} + \lambda^2|\mathbf{d}|^2 = |\mathbf{a} - \mathbf{b}|^2 + 2\lambda(\mathbf{a} - \mathbf{b}) \cdot \mathbf{d} + \lambda^2$.
This is a quadratic equation in λ with positive leading coefficient, so it has a minimum.
Differentiating: $\frac{d}{d\lambda}(f^2) = 2(\mathbf{a} - \mathbf{b}) \cdot \mathbf{d} + 2\lambda = 0$.
Therefore: $\lambda_0 = -(\mathbf{a} - \mathbf{b}) \cdot \mathbf{d} = (\mathbf{b} - \mathbf{a}) \cdot \mathbf{d}$.
(ii) The vector $\overrightarrow{PB} = \mathbf{b} - (\mathbf{a} + \lambda_0\mathbf{d}) = \mathbf{b} - \mathbf{a} - \lambda_0\mathbf{d}$.
Taking dot product with $\mathbf{d}$:

$$\overrightarrow{PB} \cdot \mathbf{d} = (\mathbf{b} - \mathbf{a}) \cdot \mathbf{d} - \lambda_0 = (\mathbf{b} - \mathbf{a}) \cdot \mathbf{d} - (\mathbf{b} - \mathbf{a}) \cdot \mathbf{d} = 0$$

Therefore PB is perpendicular to the direction of line l.
(iii) If B is on the sphere closest to l, then OBP is a straight line, so $\mathbf{b}$ is parallel to $\mathbf{a} + \lambda_0\mathbf{d}$.
The shortest distance is $|\overrightarrow{PB}| = |\mathbf{b} - \mathbf{a} - \lambda_0\mathbf{d}|$.
Since $\mathbf{b}$ is on the unit sphere, $|\mathbf{b}| = 1$. Using the perpendicularity and the assumption:

$$\text{Shortest distance} = |\mathbf{b} - \mathbf{a} - \lambda_0\mathbf{d}| = \sqrt{|\mathbf{a}|^2 - (\mathbf{a} \cdot \mathbf{d})^2} - 1$$

if the sphere and line don't intersect, or 0 if they do.

Takeaways 3.7

- Minimize $f^2(\lambda)$ by setting derivative to zero
- Minimum distance occurs when connecting vector is perpendicular to line direction
- Geometric interpretation: shortest distance from line to sphere

3.8 Solution to Problem 8: Triangle Inequality and Rectangular Prism

Solution 3.8

(i) A rectangular prism with edge lengths a, b, c has surface area

$$S = 2(ab + bc + ca).$$

The three face areas are ab, bc, and ca. Their arithmetic mean is

$$A = \frac{ab + bc + ca}{3} = \frac{S}{6}.$$

The given inequality states that for three positive numbers with arithmetic mean A,

$$\frac{(ab)(bc)(ca)}{A^3} \le 1.$$

Substituting $A = S/6$,

$$\frac{a^2b^2c^2}{(S/6)^3} \le 1 \quad \Longrightarrow \quad a^2b^2c^2 \le \left(\frac{S}{6}\right)^3.$$

Taking square roots (all quantities are positive),

$$abc \le \left(\frac{S}{6}\right)^{3/2}.$$

(ii) Equality in the given inequality occurs when $ab = bc = ca$. Since $a, b, c > 0$, this forces $a = b = c$. When $a = b = c$, the prism is a cube. Equality in the inequality means the left-hand side is maximised, so the volume abc is largest when $a = b = c$. Therefore, for fixed surface area S, the cube has maximum volume.

Takeaways 3.8

- Surface area: express S in terms of the three face areas ab, bc, ca
- Given inequality: applied with arithmetic mean $A = S/6$
- Equality: all face areas equal $\implies a = b = c$, giving maximum volume

3.9 Solution to Problem 9: Three Unit Vectors Optimization

Solution 3.9

Let θ be the angle between $\mathbf{a}$ and $\mathbf{a}+\mathbf{b}+\mathbf{c}$. Then

$$\cos\theta = \frac{\mathbf{a}\cdot(\mathbf{a}+\mathbf{b}+\mathbf{c})}{|\mathbf{a}|\,|\mathbf{a}+\mathbf{b}+\mathbf{c}|} = \frac{1+\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}}{|\mathbf{a}+\mathbf{b}+\mathbf{c}|}.$$

Since $\mathbf{a}\perp\mathbf{b}$, we have $\mathbf{a}\cdot\mathbf{b}=0$. Write $\mathbf{a}\cdot\mathbf{c}=x$.
Because $\mathbf{b}\perp\mathbf{c}$, we have $\mathbf{b}\cdot\mathbf{c}=0$, so

$$|\mathbf{a}+\mathbf{b}+\mathbf{c}|^2 = |\mathbf{a}|^2+|\mathbf{b}|^2+|\mathbf{c}|^2+2(\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}+\mathbf{b}\cdot\mathbf{c}) = 3+2x.$$

Hence $\cos\theta = \dfrac{1+x}{\sqrt{3+2x}}$.
To minimise θ, maximise $\cos\theta$. Differentiating,

$$\frac{d}{dx}\left(\frac{1+x}{\sqrt{3+2x}}\right) = \frac{\sqrt{3+2x}-(1+x)/\sqrt{3+2x}}{3+2x} = \frac{2+x}{(3+2x)^{3/2}} > 0 \quad \text{for } x > -2.$$

So $\cos\theta$ is increasing in x on the relevant domain. By Cauchy–Schwarz, $|\mathbf{a}\cdot\mathbf{c}|\le 1$, and by symmetry the optimum occurs when $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form an orthonormal set, giving $x=0$ and

$$\cos\theta = \frac{1}{\sqrt{3}}.$$

Therefore the answer is **(B)**.

Takeaways 3.9

- Dot product: express $\cos\theta$ using $|\mathbf{a}+\mathbf{b}+\mathbf{c}|^2$
- Constraints: $\mathbf{a}\perp\mathbf{b}$ and $\mathbf{b}\perp\mathbf{c}$ simplify the expansion
- Optimisation: differentiate $\cos\theta$ as a function of $x=\mathbf{a}\cdot\mathbf{c}$

3.10 Solution to Problem 10: Complex Numbers with Argument Condition

Solution 3.10

We are given $\arg(z/w) = -\pi/2$, so z/w is a negative purely imaginary number. Write

$$\frac{z}{w} = -ik \qquad \text{for some real } k > 0,$$

giving $z = -ikw$. Then

$$\begin{aligned}\left|\frac{z-w}{z+w}\right| &= \left|\frac{-ikw-w}{-ikw+w}\right| = \left|\frac{-w(1+ik)}{w(1-ik)}\right| \\ &= \left|\frac{1+ik}{1-ik}\right| = \frac{|1+ik|}{|1-ik|} = \frac{\sqrt{1+k^2}}{\sqrt{1+k^2}} = 1.\end{aligned}$$

Alternatively, since z/w is purely imaginary, z and w are perpendicular in the Argand plane. The points $z-w$ and $z+w$ are diagonals of a rectangle with sides parallel to the real and imaginary directions, so they have equal length and the ratio is 1.

Therefore $\left|\dfrac{z-w}{z+w}\right| = 1$.

Takeaways 3.10

- $\arg(z/w) = -\pi/2$ means z/w is negative purely imaginary
- Write $z = -ikw$ with real $k > 0$ and simplify the modulus
- Geometric view: numerator and denominator are equal diagonals of a rectangle

Remark 3.1

An easy solution is geometric: the numerator and denominator are diagonals of a rectangle.

3.11 Solution to Problem 11: Vectors and Complex Numbers

Solution 3.11

(i) The midpoint M of AB has position vector

$$\vec{OM} = \frac{1}{2}(\mathbf{a}+\mathbf{b}) = \mathbf{a} + \frac{1}{2}(\mathbf{b}-\mathbf{a}).$$

This is a convex combination of $\mathbf{a}$ and $\mathbf{b}$, so M lies on the line segment AB (it equals $\mathbf{a}$ when the parameter is 0 and $\mathbf{b}$ when the parameter is 1).

(ii) The centroid G has position vector

$$\vec{OG} = \frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}) = \frac{2}{3}\cdot\frac{1}{2}(\mathbf{a}+\mathbf{b}) + \frac{1}{3}\mathbf{c} = \frac{2}{3}\vec{OM} + \frac{1}{3}\mathbf{c}.$$

Since the coefficients $\frac{2}{3}$ and $\frac{1}{3}$ are positive and sum to 1, G lies on the line segment MC, between M and C.

(iii) Suppose, for contradiction, that $\frac{1}{3}(x+w+z)$ is a cube root of xwz. Then

$$\left|\frac{1}{3}(x+w+z)\right| = \left|(xwz)^{1/3}\right| = 1,$$

because $|x| = |w| = |z| = 1$. By the triangle inequality,

$$\left|\frac{1}{3}(x+w+z)\right| \leq \frac{1}{3}(|x|+|w|+|z|) = 1.$$

Equality holds only when x, w, and z have the same argument, so they are positive scalar multiples of one another. With equal moduli, this means $x = w = z$, contradicting that the three numbers are all different.

Therefore $\frac{1}{3}(x+w+z)$ is never a cube root of xwz.

Takeaways 3.11

- Midpoint: $\vec{OM} = \frac{1}{2}(\mathbf{a}+\mathbf{b})$ lies on segment AB
- Centroid: $\vec{OG} = \frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$ lies on segment MC
- Triangle inequality: equality requires equal arguments, hence equal complex numbers

3.12 Solution to Problem 12: Bar Magnet and Falling Object

Solution 3.12

(i) Using $a = v\frac{dv}{dx}$:

$$v\frac{dv}{dx} = \frac{27g}{x^3} - g = g\left(\frac{27}{x^3} - 1\right)$$

Separating variables and integrating:

$$\int_0^v v\,dv = g\int_6^x \left(\frac{27}{x^3} - 1\right)dx$$

$$\frac{v^2}{2} = g\left[-\frac{27}{2x^2} - x\right]_6^x = g\left(-\frac{27}{2x^2} - x + \frac{27}{2\cdot 36} + 6\right)$$

$$= g\left(-\frac{27}{2x^2} - x + \frac{3}{8} + 6\right) = g\left(\frac{51}{8} - x - \frac{27}{2x^2}\right)$$

Therefore: $v^2 = g\left(\frac{51}{4} - 2x - \frac{27}{x^2}\right)$.
(ii) When the object comes to rest, $v = 0$:

$$\frac{51}{4} - 2x - \frac{27}{x^2} = 0$$

Multiplying by $4x^2$: $51x^2 - 8x^3 - 108 = 0$, or $8x^3 - 51x^2 + 108 = 0$.
We need to find the root where $x < 6$ (since object falls downward). Trying values: - $x = 1.5$: $8(3.375) - 51(2.25) + 108 = 27 - 114.75 + 108 = 20.25$ - $x = 2$: $8(8) - 51(4) + 108 = 64 - 204 + 108 = -32$ - $x = 1.7$: $8(4.913) - 51(2.89) + 108 = 39.304 - 147.39 + 108 = -0.086$
By intermediate value theorem and refinement, $x \approx 1.7$ m (to 1 decimal place).

Takeaways 3.12

- Use $a = v\frac{dv}{dx}$ for position-dependent acceleration
- Integrate to find velocity as function of position
- Solve $v = 0$ to find rest positions

3.13 Solution to Problem 13: Circle and Cosine Function

Solution 3.13

At the intersection point $P(a, b)$ in the first quadrant, the point lies on $y = \cos(kx)$ and on a circle centred at the origin, so

$$b = \cos(ka), \qquad a^2 + b^2 = r^2$$

for some radius r.
The slope of the radius OP is b/a. The slope of the tangent to $y = \cos(kx)$ at $x = a$ is $-k\sin(ka)$. Since the radius is perpendicular to the tangent,

$$\frac{b}{a} \cdot \big(-k\sin(ka)\big) = -1 \quad \Longrightarrow \quad kb\sin(ka) = a.$$

Substituting $b = \cos(ka)$,

$$k\cos(ka)\sin(ka) = a \quad \Longrightarrow \quad \frac{k}{2}\sin(2ka) = a.$$

Also $r^2 = a^2 + \cos^2(ka)$.
The circle has exactly two intersections with the graph and is never above it, so it is tangent at P in the first quadrant. For this configuration to occur, the cosine curve must complete at least one full oscillation on the interval where the circle meets the axis, which requires period $T = 2\pi/k < 2\pi$, i.e. $k > 1$. When $k \leq 1$, the geometry of the graph and the tangency condition cannot produce exactly two intersections with the circle always below the curve.
Therefore $k > 1$.

Takeaways 3.13

- Perpendicularity: product of slopes equals -1
- Use $b = \cos(ka)$ and the double-angle identity to relate a and k
- Geometry: tangency with exactly two intersections requires $k > 1$

3.14 Solution to Problem 14: Cube Roots of Unity and Trigonometric Products

Solution 3.14

(i) Since $w = e^{2\pi i/3}$ and $\gamma^3 = w$, we have $\gamma^9 = w^3 = 1$, so γ is a 9th root of unity.
Write $\gamma = e^{2\pi ik/9}$ for some integer k. Then

$$\gamma + \overline{\gamma} = 2\cos\left(\frac{2\pi k}{9}\right).$$

From $\gamma^3 = e^{2\pi ik/3} = w = e^{2\pi i/3}$ we need $k \equiv 1 \pmod 3$, so $k = 1, 4, 7$.
Take $k = 1$, so $z = \gamma + \overline{\gamma} = 2\cos(2\pi/9)$. Using the triple-angle formula $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$ with $\theta = 2\pi/9$,

$$\cos\left(\frac{6\pi}{9}\right) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} = 4\cos^3\left(\frac{2\pi}{9}\right) - 3\cos\left(\frac{2\pi}{9}\right).$$

With $z = 2\cos(2\pi/9)$, this becomes $-\frac{1}{2} = \frac{z^3}{8} - \frac{3z}{2}$, so $z^3 - 3z + 1 = 0$. Hence $\gamma + \overline{\gamma}$ is a real root of $z^3 - 3z + 1 = 0$.
(ii) From part (i), the roots of $z^3 - 3z + 1 = 0$ include $2\cos(2\pi/9)$, $2\cos(4\pi/9)$, and $2\cos(8\pi/9)$. Using Vieta's formulas (or a standard product identity),

$$\cos\frac{2\pi}{9}\cos\frac{4\pi}{9}\cos\frac{8\pi}{9} = \frac{1}{8}.$$

For $n \geq 1$, the angles $2^n\pi/9$, $2^{n+1}\pi/9$, $2^{n+2}\pi/9$ depend on $2^n \bmod 9$. Since powers of 2 modulo 9 cycle through $2, 4, 8, 7, 5, 1$, the triple of angles cycles through the same set, so

$$\cos\frac{2^n\pi}{9}\cos\frac{2^{n+1}\pi}{9}\cos\frac{2^{n+2}\pi}{9} = \frac{1}{8} \quad \text{for all integers } n \geq 1.$$

Takeaways 3.14

- If $\gamma^3 = w$, then γ is a 9th root of unity with $k \equiv 1 \pmod 3$
- Triple-angle formula: links $\cos(2\pi/9)$ to the cubic $z^3 - 3z + 1 = 0$
- Powers of 2 modulo 9 cycle, so the product in part (ii) is always $1/8$

3.15 Solution to Problem 15: Integer Equation with Large Exponents

Solution 3.15

Parity-only argument (no modular arithmetic). Consider even and odd values of n separately.
If n is even, then $n + 1$ is odd, so $(n + 1)^{41}$ is odd; also n^{40} is even, so $79n^{40}$ is even. Thus the left-hand side is odd minus even, which is **odd**.
If n is odd, then $n + 1$ is even, so $(n + 1)^{41}$ is even; also n^{40} is odd, so $79n^{40}$ is odd. Thus the left-hand side is even minus odd, which is again **odd**.
In every case the left-hand side is odd, while the right-hand side is 2, which is even. An odd integer cannot equal an even integer, so there is no integer n satisfying the equation.
Note. Modular arithmetic is not in the syllabus, but it is a useful tool. For completeness, the same argument in mod 2 is: in both cases above, $(n + 1)^{41} - 79n^{40} \equiv 1 \pmod 2$, whereas $2 \equiv 0 \pmod 2$.

Takeaways 3.15

- Parity alone shows the left side is always odd while 2 is even, so no solution exists
- Split into even n and odd n; powers preserve parity in each case
- Modulo 2 language is optional here but gives the same contradiction compactly

3.16 Solution to Problem 16: Projectile with Quadratic Resistance

Solution 3.16

(i) On the upward journey, the acceleration is $a = -g - kv^2$ (negative because both gravity and resistance oppose motion).
Using $a = v\frac{dv}{dx}$:

$$v\frac{dv}{dx} = -g - kv^2$$

Separating variables:

$$\frac{v}{g + kv^2}\,dv = -dx$$

Integrating from $v = v_0$ at $x = 0$ to $v = 0$ at $x = H$:

$$\int_{v_0}^{0} \frac{v}{g + kv^2}\,dv = -\int_0^H dx$$

For the left integral, let $u = g + kv^2$, so $du = 2kv\,dv$, giving:

$$\int_{v_0}^{0} \frac{v}{g + kv^2}\,dv = \frac{1}{2k}\int_{g+kv_0^2}^{g} \frac{du}{u} = \frac{1}{2k}\ln\left(\frac{g}{g + kv_0^2}\right) = -\frac{1}{2k}\ln\left(\frac{g + kv_0^2}{g}\right)$$

Therefore: $-H = -\frac{1}{2k}\ln\left(\frac{g+kv_0^2}{g}\right)$, so

$$H = \frac{1}{2k}\ln\left(\frac{kv_0^2 + g}{g}\right).$$

(ii) On the downward journey, acceleration is $a = g - kv^2$ (gravity assists, resistance opposes).
Using the same technique: $v\frac{dv}{dx} = g - kv^2$.
Integrating from $v = 0$ at $x = H$ to $v = v_1$ at $x = 0$:

$$\int_0^{v_1} \frac{v}{g - kv^2}\,dv = \int_H^0 dx = -H$$

Let $u = g - kv^2$, so $du = -2kv\,dv$:

$$\int_0^{v_1} \frac{v}{g - kv^2}\,dv = -\frac{1}{2k}\int_g^{g-kv_1^2} \frac{du}{u} = -\frac{1}{2k}\ln\left(\frac{g - kv_1^2}{g}\right) = \frac{1}{2k}\ln\left(\frac{g}{g - kv_1^2}\right)$$

So: $\frac{1}{2k}\ln\left(\frac{g}{g-kv_1^2}\right) = -H = -\frac{1}{2k}\ln\left(\frac{kv_0^2+g}{g}\right)$.
Therefore: $\ln\left(\frac{g}{g-kv_1^2}\right) = -\ln\left(\frac{kv_0^2+g}{g}\right)$, so $\frac{g}{g-kv_1^2} = \frac{g}{kv_0^2+g}$.
Cross-multiplying: $(g - kv_1^2)(kv_0^2 + g) = g^2$.
Expanding: $g(kv_0^2 + g) - kv_1^2(kv_0^2 + g) = g^2$, so $gkv_0^2 + g^2 - kv_0^2v_1^2 - kgv_1^2 = g^2$.
Simplifying: $gkv_0^2 - kv_0^2v_1^2 - kgv_1^2 = 0$, so $gk(v_0^2 - v_1^2) = kv_0^2v_1^2$.
Therefore: $g(v_0^2 - v_1^2) = kv_0^2v_1^2$.

Takeaways 3.16

- Quadratic resistance: Use $a = v\frac{dv}{dx}$ for position-dependent acceleration
- Integration: Use substitution $u = g \pm kv^2$ to integrate $\frac{v}{g \pm kv^2}$
- Energy approach: Can also use work-energy theorem for part (ii)

3.17 Solution to Problem 17: Integration with Recurrence Relations

Solution 3.17

(i) Using integration by parts with $u = \sin^{n-1}\theta$ and $dv = \sin\theta\, d\theta$:

$$J_n = \int_0^{\pi/2} \sin^n\theta\, d\theta = \int_0^{\pi/2} \sin^{n-1}\theta \cdot \sin\theta\, d\theta$$

Let $u = \sin^{n-1}\theta$, $dv = \sin\theta\, d\theta$, so $du = (n-1)\sin^{n-2}\theta\cos\theta\, d\theta$ and $v = -\cos\theta$:

$$J_n = \left[-\sin^{n-1}\theta\cos\theta\right]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \sin^{n-2}\theta\cos^2\theta\, d\theta$$

The boundary term is 0. Using $\cos^2\theta = 1 - \sin^2\theta$:

$$J_n = (n-1)\int_0^{\pi/2} \sin^{n-2}\theta(1-\sin^2\theta)\, d\theta = (n-1)(J_{n-2} - J_n)$$

Solving: $J_n = (n-1)J_{n-2} - (n-1)J_n$, so $nJ_n = (n-1)J_{n-2}$.
Therefore: $J_n = \frac{n-1}{n}J_{n-2}$.
(ii) Using substitution $x = \sin^2\theta$, so $dx = 2\sin\theta\cos\theta\, d\theta$ and $1 - x = \cos^2\theta$:

$$I_n = \int_0^1 x^n(1-x)^n\, dx = \int_0^{\pi/2} \sin^{2n}\theta\cos^{2n}\theta \cdot 2\sin\theta\cos\theta\, d\theta$$

$$= 2\int_0^{\pi/2} \sin^{2n+1}\theta\cos^{2n+1}\theta\, d\theta = 2\int_0^{\pi/2} (\sin\theta\cos\theta)^{2n+1}\, d\theta$$

Using $\sin(2\theta) = 2\sin\theta\cos\theta$, so $\sin\theta\cos\theta = \frac{1}{2}\sin(2\theta)$:

$$I_n = 2\int_0^{\pi/2} \left(\frac{1}{2}\sin(2\theta)\right)^{2n+1} d\theta = \frac{2}{2^{2n+1}}\int_0^{\pi/2} \sin^{2n+1}(2\theta)\, d\theta$$

Let $u = 2\theta$, so $du = 2d\theta$:

$$I_n = \frac{1}{2^{2n+1}}\int_0^{\pi} \sin^{2n+1} u \cdot \frac{du}{2} = \frac{1}{2^{2n+2}}\int_0^{\pi} \sin^{2n+1} u\, du$$

By symmetry: $\int_0^\pi \sin^{2n+1} u\, du = 2\int_0^{\pi/2} \sin^{2n+1} u\, du$.
Therefore: $I_n = \frac{1}{2^{2n+2}} \cdot 2J_{2n+1} = \frac{1}{2^{2n+1}}J_{2n+1}$.
(iii) From part (ii): $I_n = \frac{1}{2^{2n+1}}J_{2n+1}$ and $I_{n-1} = \frac{1}{2^{2n-1}}J_{2n-1}$.
From part (i): $J_{2n+1} = \frac{2n}{2n+1}J_{2n-1}$.
Therefore:

$$I_n = \frac{1}{2^{2n+1}} \cdot \frac{2n}{2n+1}J_{2n-1} = \frac{2n}{2n+1} \cdot \frac{1}{2^{2n+1}}J_{2n-1} = \frac{2n}{2n+1} \cdot \frac{1}{4} \cdot \frac{1}{2^{2n-1}}J_{2n-1}$$

$$= \frac{n}{2(2n+1)} \cdot I_{n-1} = \frac{n}{4n+2}I_{n-1}$$

Takeaways 3.17

- Integration by parts: Use to derive recurrence relations for powers of trigonometric functions
- Substitution: $x = \sin^2\theta$ converts polynomial integrals to trigonometric integrals
- Combining recurrences: Use multiple recurrence relations together to prove new results

3.18 Solution to Problem 18: Curve on a Sphere

Solution 3.18

The curve lies on the sphere $x^2 + y^2 + z^2 = 9$ of radius 3. The given functions suggest using z as the main parameter.
Let $t = z/3$, so $t \in [-1, 1]$ as z runs from -3 to 3. The curve spirals three times around the z-axis, so the azimuthal angle should increase by 6π as t goes from -1 to 1. Take

$$\theta = 3\pi t.$$

At fixed height $z = 3t$, a point on the sphere has horizontal radius $\sqrt{9-z^2} = 3\sqrt{1-t^2}$.
Using the given functions $g(x) = \sqrt{9-x^2}$ and $f(x) = \cos(\pi x)$ with $x = z/3 = t$, a parameterisation is

$$x(t) = 3\sqrt{1-t^2}\cos(3\pi t), \qquad y(t) = 3\sqrt{1-t^2}\sin(3\pi t), \qquad z(t) = 3t, \qquad t \in [-1, 1].$$

Check that the curve stays on the sphere:

$$\begin{aligned} x^2 + y^2 + z^2 &= 9(1-t^2)\big(\cos^2(3\pi t) + \sin^2(3\pi t)\big) + 9t^2 \\ &= 9(1-t^2) + 9t^2 = 9. \end{aligned}$$

Takeaways 3.18

- Sphere constraint: $x^2 + y^2 + z^2 = 9$ with radius 3
- Parameter $t = z/3$ makes the height vary linearly from -3 to 3
- Spiralling: use $\theta = 3\pi t$ and horizontal radius $3\sqrt{1-t^2}$

3.19 Solution to Problem 19: Complex Numbers and Equilateral Triangles

Solution 3.19

(i) Since $w = e^{2\pi i/3}$, we have $w^3 = e^{2\pi i} = 1$, so w is a non-trivial cube root of unity. For $w \neq 1$,

$$1 + w + w^2 = \frac{1 - w^3}{1 - w} = \frac{1 - 1}{1 - w} = 0.$$

(ii) If triangle ABC is anticlockwise and equilateral, a rotation of $120°$ about the centroid $G = \dfrac{a + b + c}{3}$ maps each vertex to the next:

$$b - G = w(a - G), \qquad c - G = w(b - G).$$

From the first equation,

$$b = G + w(a - G) = (1 - w)G + wa = \frac{1 - w}{3}(a + b + c) + wa.$$

Rearranging and using $1 + w + w^2 = 0$ (so $1 - w = -w - w^2$), we obtain

$$a + bw + cw^2 = 0.$$

(iii) For an equilateral triangle, either the anticlockwise condition $a + bw + cw^2 = 0$ or the clockwise condition $a + bw^2 + cw = 0$ holds.
Multiplying $a + bw + cw^2 = 0$ by w gives $aw + bw^2 + c = 0$, so $c = -aw - bw^2$. Substituting into $a + bw^2 + cw = 0$:

$$a + bw^2 + w(-aw - bw^2) = a(1 - w^2) + bw^2(1 - w) = 0.$$

Using $1 + w + w^2 = 0$, we have $1 - w^2 = w$ and $1 - w = w^2$, so this becomes $w(a + b) = 0$, hence $a + b = -c$ and $a + b + c = 0$.
Subtracting $a + b + c = 0$ from $a + bw + cw^2 = 0$:

$$b(w - 1) + c(w^2 - 1) = 0.$$

Since $w^2 - 1 = (w - 1)(w + 1)$ and $w + 1 = -w^2$, we get $(w - 1)(b - cw^2) = 0$. As $w \neq 1$, it follows that $b = cw^2$; similarly $a = bw^2$ and $c = aw^2$.
From $a + bw + cw^2 = 0$ and $a + b + c = 0$, expand a^2, b^2, c^2 in terms of one another and add. Using $w + w^2 = -1$,

$$a^2 + b^2 + c^2 = (a^2 + b^2 + c^2)(w + w^2) + 2(ab + bc + ca) = -(a^2 + b^2 + c^2) + 2(ab + bc + ca).$$

Therefore $2(a^2 + b^2 + c^2) = 2(ab + bc + ca)$, so

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

Takeaways 3.19

- Cube roots of unity: $1 + w + w^2 = 0$ where $w = e^{2\pi i/3}$
- Rotation by $120°$ about the centroid maps vertices of an equilateral triangle
- Use both orientations to derive $a + b + c = 0$, then simplify to the required identity

3.20 Solution to Problem 20: Inequalities with Exponentials and Factorials

Solution 3.20

(i) Define $f(x) = x - \ln x$ for $x > 0$. Then

$$f'(x) = 1 - \frac{1}{x} = \frac{x-1}{x}.$$

So $f'(x) = 0$ when $x = 1$, $f'(x) < 0$ for $0 < x < 1$, and $f'(x) > 0$ for $x > 1$. Thus f has a global minimum at $x = 1$, and

$$f(1) = 1 - \ln 1 = 1 > 0.$$

Since $f(x) > 0$ for all $x > 0$, we have $x > \ln x$ for all $x > 0$.

(ii) Apply the inequality from part (i) with $x = 1, 2, \ldots, n$:

$$1 > \ln 1, \quad 2 > \ln 2, \quad \ldots, \quad n > \ln n.$$

Adding these n inequalities,

$$1 + 2 + \cdots + n > \ln 1 + \ln 2 + \cdots + \ln n = \ln(n!).$$

Using $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2} = \dfrac{n^2+n}{2}$,

$$\frac{n^2+n}{2} > \ln(n!).$$

Since the exponential function is increasing, exponentiating both sides gives

$$e^{(n^2+n)/2} > n! \quad \Longrightarrow \quad e^{n^2+n} > (n!)^2.$$

Takeaways 3.20

- Function analysis: Find the minimum of $f(x) = x - \ln x$ to prove a basic inequality
- Sum of inequalities: Add the same inequality for $x = 1, 2, \ldots, n$
- Factorial bounds: Exponentiate to relate $(n!)^2$ to an exponential

3.21 Solution to Problem 21: Complex Numbers and Region Sketching

Solution 3.21

Note: Exponential form is being removed from the 2024 syllabus; this solution stays in modulus–argument (trigonometric) form, though the shorthand is still handy to know.
First simplify the expression:

$$\frac{xz + yw}{z} = x + y\,\frac{w}{z}.$$

Since $|w| = |z| = 1$, we have $\left|\frac{w}{z}\right| = 1$. The given condition $\operatorname{Arg}\left(\frac{z}{w}\right) \in \left(\frac{\pi}{2}, \pi\right)$ means

$$\operatorname{Arg}\left(\frac{w}{z}\right) = -\operatorname{Arg}\left(\frac{z}{w}\right) \in \left(-\pi, -\frac{\pi}{2}\right).$$

Write $\alpha = \operatorname{Arg}\left(\frac{w}{z}\right)$, so $-\pi < \alpha < -\frac{\pi}{2}$ and $\frac{w}{z} = \cos\alpha + i\sin\alpha$ lies in the third quadrant.
We want

$$\operatorname{Arg}\left(x + y\,\frac{w}{z}\right) \in \left(\frac{\pi}{2}, \pi\right),$$

so $x + y\frac{w}{z}$ must lie in the second quadrant. Expanding,

$$x + y\,\frac{w}{z} = (x + y\cos\alpha) + i(y\sin\alpha).$$

For a point in the second quadrant, the real part is negative and the imaginary part is positive:

$$x + y\cos\alpha < 0, \qquad y\sin\alpha > 0.$$

Since $\alpha \in \left(-\pi, -\frac{\pi}{2}\right)$, we have $\sin\alpha < 0$, so $y\sin\alpha > 0$ requires $y < 0$. Also $\cos\alpha < 0$, so the boundary line $x + y\cos\alpha = 0$ has negative slope.
The required region is therefore

$$\{(x, y) : y < 0 \text{ and } x + y\cos\alpha < 0\},$$

where $\alpha = \operatorname{Arg}\left(\frac{w}{z}\right)$ is fixed by the given data. This is the half-plane in the lower half of the xy-plane below the line through the origin with equation $x = -y\cos\alpha$.

Takeaways 3.21

- Simplify first: $\dfrac{xz + yw}{z} = x + y\dfrac{w}{z}$
- Argument conditions: Use $\operatorname{Arg}(w/z) = -\operatorname{Arg}(z/w)$ to fix the quadrant of w/z
- Region sketching: Second quadrant means negative real part and positive imaginary part

3.22 Solution to Problem 22: Mechanics with Ropes and Forces

Solution 3.22

(i) At point P the machine is in equilibrium. Resolve the tensions into horizontal and vertical components. Horizontally, the components balance:

$$T_1 \cos\theta = T_2 \cos\phi. \tag{1}$$

Vertically, the upward components support the weight:

$$T_1 \sin\theta + T_2 \sin\phi = Mg. \tag{2}$$

From (1), $T_1 = \dfrac{T_2 \cos\phi}{\cos\theta}$. Substituting into (2),

$$\frac{T_2 \cos\phi}{\cos\theta} \sin\theta + T_2 \sin\phi = Mg,$$

$$T_2(\cos\phi \tan\theta + \sin\phi) = Mg.$$

Dividing by $T_2 \cos\phi$,

$$\tan\theta + \frac{\sin\phi}{\cos\phi} = \frac{Mg}{T_2 \cos\phi} \quad \Longrightarrow \quad \tan\theta = \tan\phi + \frac{Mg}{T_2 \cos\phi}.$$

(ii) From part (i), since $Mg > 0$, $T_2 > 0$, and $\cos\phi > 0$ for the configuration shown, the term $\dfrac{Mg}{T_2 \cos\phi}$ is positive. Hence

$$\tan\theta > \tan\phi \quad \Longrightarrow \quad \theta > \phi.$$

Now suppose P could be lifted to a height $\dfrac{2h}{3}$ above the floor. Using the fixed horizontal distances from P to the vertical lines through C and F, the geometry of the diagram expresses $\tan\theta$ and $\tan\phi$ in terms of the vertical distances $\dfrac{h}{3}$ (to the ceiling) and $\dfrac{2h}{3}$ (to the floor). At this height the resulting ratio of $\tan\theta$ to $\tan\phi$ would violate the strict inequality $\tan\theta > \tan\phi + \dfrac{Mg}{T_2 \cos\phi}$ required by part (i). This contradiction shows that P cannot be lifted to $\dfrac{2h}{3}$ metres above the floor.

Takeaways 3.22

- Force resolution: Resolve each tension into horizontal and vertical components at P
- Static equilibrium: The sum of forces in each direction is zero
- Geometric constraints: Combine the force relation with height ratios to rule out a position

3.23 Solution to Problem 23: Simple Harmonic Motion

Solution 3.23

The particle oscillates between 0.05 m and 0.17 m, so the amplitude is half the total range:

$$A = \frac{0.17 - 0.05}{2} = 0.06 \text{ m}.$$

The frequency is $f = 40$ Hz, so the angular frequency is

$$\omega = 2\pi f = 80\pi \text{ rad/s}.$$

For simple harmonic motion, the maximum acceleration occurs at the endpoints and has magnitude

$$a_{\max} = \omega^2 A = (80\pi)^2 \times 0.06 = 6400\pi^2 \times 0.06 = 384\pi^2 \text{ m/s}^2.$$

The maximum force on the mass is $F_{\max} = ma_{\max}$, so with $m = 0.8$ kg,

$$F_{\max} = 0.8 \times 384\pi^2 = 307.2\pi^2 \approx 3032 \text{ N}.$$

To the nearest newton, $F_{\max} \approx 3032$ N.

Takeaways 3.23

- SHM amplitude: $A = \dfrac{\text{max displacement} - \text{min displacement}}{2}$
- Maximum acceleration: $a_{\max} = \omega^2 A$ where $\omega = 2\pi f$
- Newton's second law: $F_{\max} = ma_{\max}$ at maximum acceleration

3.24 Solution to Problem 24: Projectile with Linear Resistance

Solution 3.24

Take upward as positive. The resistive force has magnitude $0.1Mv$, opposing the motion, so while the projectile is rising

$$M\frac{dv}{dt} = -Mg - 0.1Mv \quad \Longrightarrow \quad \frac{dv}{dt} = -10 - 0.1v = -0.1(v + 100).$$

Separate variables:

$$\frac{dv}{v + 100} = -0.1\,dt.$$

Integrating and using $v = v_0$ when $t = 0$,

$$\ln(v + 100) = -0.1t + \ln(v_0 + 100) \quad \Longrightarrow \quad v = (v_0 + 100)e^{-0.1t} - 100.$$

The projectile reaches its highest point when $v = 0$:

$$0 = (v_0 + 100)e^{-0.1t_1} - 100 \quad \Longrightarrow \quad t_1 = 10\ln\left(\frac{v_0 + 100}{100}\right).$$

On the downward journey, gravity and resistance act in opposite directions on the vertical motion, giving

$$\frac{dv}{dt} = 10 - 0.1v.$$

Solving this equation and using the condition that the total time from launch to landing is 7 seconds determines v_0. Solving numerically gives

$$v_0 \approx 49.5 \text{ m/s} \quad \text{(to 1 decimal place)}.$$

Takeaways 3.24

- Linear resistance: Use $\frac{dv}{dt} = -g - kv$ on the way up and $g - kv$ on the way down
- Separable ODE: Integrate to obtain v as a function of t
- Boundary conditions: Use the apex condition $v = 0$ and total flight time 7 s to find v_0

3.25 Solution to Problem 25: Complex Numbers with Three Conditions

Solution 3.25

Since $|z_1| = |z_2| = |z_3| = r$, write

$$z_k = r(\cos\theta_k + i\sin\theta_k) \qquad (k = 1, 2, 3).$$

From $z_1 z_2 z_3 = 1$,

$$r^3\big[\cos(\theta_1 + \theta_2 + \theta_3) + i\sin(\theta_1 + \theta_2 + \theta_3)\big] = 1.$$

The right-hand side is the real number 1, so the imaginary part is zero and the modulus is 1. Hence $r^3 = 1$ (so $r = 1$) and

$$\theta_1 + \theta_2 + \theta_3 = 2\pi k \quad \text{for some integer } k.$$

Thus each z_k lies on the unit circle: $z_k = \cos\theta_k + i\sin\theta_k$.
From $z_1 + z_2 + z_3 = 1$,

$$(\cos\theta_1 + \cos\theta_2 + \cos\theta_3) + i(\sin\theta_1 + \sin\theta_2 + \sin\theta_3) = 1.$$

The imaginary parts must sum to zero. A natural symmetric choice is to take one number real and the other two a conjugate pair: let

$$z_1 = 1, \qquad z_2 = \cos\theta + i\sin\theta, \qquad z_3 = \cos\theta - i\sin\theta.$$

Then

$$z_1 + z_2 + z_3 = 1 + 2\cos\theta = 1 \quad\Longrightarrow\quad \cos\theta = 0,$$

so $\theta = \frac{\pi}{2}$ or $\frac{3\pi}{2}$.
For $\theta = \frac{\pi}{2}$: $z_1 = 1$, $z_2 = i$, $z_3 = -i$.
Check: $|1| = |i| = |-i| = 1$ ✓, $1 + i + (-i) = 1$ ✓, $1 \cdot i \cdot (-i) = 1$ ✓.
For $\theta = \frac{3\pi}{2}$: $z_2 = -i$, $z_3 = i$, giving the same set permuted.
By symmetry, any of z_1, z_2, z_3 may be the real number 1, with the other two equal to i and $-i$.
Therefore the solutions are all permutations of $\{1, i, -i\}$.

Takeaways 3.25

- Equal moduli: Write each number in modulus–argument form with a common radius r
- Product condition: $z_1 z_2 z_3 = 1$ forces $r = 1$ and a sum of arguments equal to $2\pi k$
- Symmetry: A conjugate pair together with a real number simplifies the sum condition

3.26 Solution to Problem 26: Complex Numbers and Equilateral Triangle

Solution 3.26

(i) We are given

$$z_2 = \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) z_1.$$

Multiplying by $\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$ rotates z_1 through $\frac{\pi}{3}$ about the origin, so

$$|z_2| = |z_1| \qquad \text{and} \qquad \operatorname{Arg}(z_2) = \operatorname{Arg}(z_1) + \frac{\pi}{3}.$$

Since O is the origin, $|OA| = |z_1|$, $|OB| = |z_2| = |z_1|$, and $\angle AOB = \frac{\pi}{3}$. Triangle OAB therefore has two equal sides and included angle $60°$, so it is equilateral.

(ii) Using $z_2 = \left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) z_1$, first compute

$$\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^2 = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} = -\frac{1}{2} + \frac{i\sqrt{3}}{2}.$$

Then

$$\begin{aligned} z_1^2 + z_2^2 &= z_1^2 + \left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^2 z_1^2 \\ &= z_1^2\left[1 + \left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)\right] \\ &= z_1^2\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right). \end{aligned}$$

Also,

$$z_1 z_2 = z_1 \cdot \left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) z_1 = \left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) z_1^2.$$

Hence $z_1^2 + z_2^2 = z_1 z_2$.

Takeaways 3.26

- Rotation: Multiplying by $\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$ rotates through $60°$
- Equilateral triangle: Two equal sides with a $60°$ included angle imply equilateral
- Complex algebra: Square the rotation factor using De Moivre's theorem

3.27 Solution to Problem 27: Particle Falling with Resistance

Solution 3.27

Given $a = 10(1 - (kv)^2)$ where $k = 0.01$, so $a = 10(1 - 0.0001v^2)$.
Using $a = \frac{dv}{dt}$:

$$\frac{dv}{dt} = 10(1 - 0.0001v^2) = 10 - 0.001v^2$$

The terminal velocity occurs when $a = 0$: $v_T = \sqrt{\frac{10}{0.001}} = 100$ m/s.
Separating variables:

$$\frac{dv}{1 - 0.0001v^2} = 10\,dt$$

Using partial fractions: $\frac{1}{1-0.0001v^2} = \frac{1}{2}\left(\frac{1}{1-0.01v} + \frac{1}{1+0.01v}\right)$.
Integrating from $v = 0$ at $t = 0$ to $v = v$ at $t = 5$:

$$\frac{1}{0.02}\ln\left|\frac{1 + 0.01v}{1 - 0.01v}\right| = 10t$$

At $t = 5$: $\ln\left|\frac{1+0.01v}{1-0.01v}\right| = 1$, so $\frac{1+0.01v}{1-0.01v} = e$.
Solving: $1 + 0.01v = e(1 - 0.01v)$, so $v = \frac{e-1}{0.01(e+1)} \approx 46.2$ m/s.

Takeaways 3.27

- Terminal velocity: Found when acceleration is zero
- Partial fractions: Use to integrate $\frac{1}{1-a^2v^2}$
- Hyperbolic tangent: The solution involves tanh function

3.28 Solution to Problem 28: Induction Proof for Series

Solution 3.28

We prove by induction that
$$\frac{1}{2^2}+\frac{1}{3^2}+\cdots+\frac{1}{n^2}<\frac{n-1}{n} \quad \text{for all integers } n\geq 2.$$
Base case ($n=2$)**:** The left-hand side is $\frac{1}{4}=0.25$ and the right-hand side is $\frac{1}{2}=0.5$, so $0.25<0.5$ ✓.
Inductive step: Assume the statement holds for $n=k\geq 2$:
$$\frac{1}{2^2}+\frac{1}{3^2}+\cdots+\frac{1}{k^2}<\frac{k-1}{k}.$$
For $n=k+1$ we need
$$\frac{1}{2^2}+\frac{1}{3^2}+\cdots+\frac{1}{k^2}+\frac{1}{(k+1)^2}<\frac{k}{k+1}.$$
By the inductive hypothesis, it suffices to show
$$\frac{k-1}{k}+\frac{1}{(k+1)^2}<\frac{k}{k+1}.$$
Rearranging,
$$\frac{k-1}{k}<\frac{k}{k+1}-\frac{1}{(k+1)^2}=\frac{k(k+1)-1}{(k+1)^2}=\frac{k^2+k-1}{(k+1)^2}.$$
Cross-multiplying (all denominators are positive for $k\geq 2$):
$$(k-1)(k+1)^2<k(k^2+k-1).$$
Expanding the left-hand side,
$$(k-1)(k^2+2k+1)=k^3+k^2-k-1.$$
So we need $k^3+k^2-k-1<k^3+k^2-k$, which simplifies to $-1<0$ ✓.
Therefore, by mathematical induction, the inequality holds for all $n\geq 2$.

Takeaways 3.28

- Base case: Verify the smallest value $n=2$
- Inductive step: Assume the result for k, then add the next term for $k+1$
- Algebraic manipulation: Reduce the inductive step to a simple inequality

3.29 Solution to Problem 29: Irrationality of Logarithm

Solution 3.29

Assume, for contradiction, that $\log_n(n+1)$ is rational. Then we can write
$$\log_n(n+1)=\frac{p}{q}$$
for some positive integers p and q. Converting to exponential form,
$$n^{p/q}=n+1 \quad\Longrightarrow\quad n^p=(n+1)^q.$$
Since n and $n+1$ are consecutive integers, they are coprime: no prime divides both.
Every prime factor of n^p divides n, while every prime factor of $(n+1)^q$ divides $n+1$. Because n and $n+1$ share no common prime factors, the equality $n^p=(n+1)^q$ can hold only if both sides equal 1.
But $n>1$, so $n^p\geq 2^p\geq 2>1$ and $(n+1)^q\geq 3^q\geq 3>1$. This is a contradiction.
Therefore $\log_n(n+1)$ is irrational.

Takeaways 3.29

- Proof by contradiction: Assume the logarithm is rational and write it as p/q
- Coprime property: Consecutive integers n and $n+1$ have no common prime factors
- Unique factorisation: Equal powers force the same prime factors on both sides

3.30 Solution to Problem 30: Proposition Logic

Solution 3.30

(i) We prove the implication directly. Suppose $k+1$ is divisible by 3. Then we can write

$$k+1 = 3m \qquad \text{for some integer } m,$$

so $k = 3m-1$. Substituting into k^3+1,

$$\begin{aligned} k^3+1 &= (3m-1)^3+1 \\ &= 27m^3 - 27m^2 + 9m - 1 + 1 \\ &= 27m^3 - 27m^2 + 9m \\ &= 9m(3m^2 - 3m + 1). \end{aligned}$$

The factor $9m$ is divisible by 3, so k^3+1 is divisible by 3. Hence proposition P is true.
(ii) The contrapositive of 'If A, then B' is 'If not B, then not A'. So the contrapositive of P is:

"If k^3+1 is not divisible by 3, then $k+1$ is not divisible by 3."

Since a statement and its contrapositive are logically equivalent, this is also true.
(iii) The converse of 'If A, then B' is 'If B, then A'. So the converse of P is:

"If k^3+1 is divisible by 3, then $k+1$ is divisible by 3."

To decide whether it is true, work modulo 3. If k^3+1 is divisible by 3, then

$$k^3 \equiv -1 \equiv 2 \pmod 3.$$

The possible residues of cubes modulo 3 are

$$0^3 \equiv 0, \qquad 1^3 \equiv 1, \qquad 2^3 \equiv 8 \equiv 2 \pmod 3.$$

So $k^3 \equiv 2 \pmod 3$ forces $k \equiv 2 \pmod 3$, and therefore

$$k+1 \equiv 2+1 \equiv 0 \pmod 3.$$

Thus $k+1$ is divisible by 3, and the converse is **true**.

Takeaways 3.30

- Direct proof: Substitute $k = 3m-1$ and expand to show divisibility by 3
- Contrapositive: Negate both the hypothesis and the conclusion
- Converse: Swap hypothesis and conclusion; modular arithmetic checks divisibility

3.31 Solution to Problem 31: Vectors and Ratios

Solution 3.31

(i) Given $\frac{CB}{AC} = \frac{m}{n}$, and $AC + CB = AB$, we have:

$$AC = \frac{n}{m+n}AB = \frac{n}{m+n}(\mathbf{b} - \mathbf{a})$$

(ii) Since $\vec{OC} = \vec{OA} + \vec{AC}$:

$$\vec{OC} = \mathbf{a} + \frac{n}{m+n}(\mathbf{b} - \mathbf{a}) = \mathbf{a} + \frac{n}{m+n}\mathbf{b} - \frac{n}{m+n}\mathbf{a}$$
$$= \left(1 - \frac{n}{m+n}\right)\mathbf{a} + \frac{n}{m+n}\mathbf{b} = \frac{m}{m+n}\mathbf{a} + \frac{n}{m+n}\mathbf{b}$$

Takeaways 3.31

- Section formula: Point dividing AB in ratio $m : n$ has position vector $\frac{m\mathbf{b}+n\mathbf{a}}{m+n}$
- Vector addition: $\vec{OC} = \vec{OA} + \vec{AC}$
- Ratio relationships: Use $AC + CB = AB$ to find lengths

3.32 Solution to Problem 32: Parallelogram Geometry

Solution 3.32

(i) We work from O and express every position vector in terms of $\mathbf{p}$ and $\mathbf{r}$.
Since $OPQR$ is a parallelogram, opposite sides are equal and parallel, so

$$\vec{OQ} = \vec{OP} + \vec{OR} = \mathbf{p} + \mathbf{r}.$$

Next, S is the midpoint of QR. Starting from R and moving halfway along $\vec{RQ}$,

$$\vec{OS} = \vec{OR} + \frac{1}{2}\vec{RQ} = \mathbf{r} + \frac{1}{2}(\vec{OQ} - \vec{OR}) = \mathbf{r} + \frac{1}{2}(\mathbf{p} + \mathbf{r} - \mathbf{r}) = \mathbf{r} + \frac{1}{2}\mathbf{p}.$$

Point T lies on both PR and OS, so we write $\vec{OT}$ in two parametric forms and equate them.
On line PR, any point from R towards P has the form

$$\vec{OT} = \vec{OR} + \lambda(\vec{OP} - \vec{OR}) = (1-\lambda)\mathbf{r} + \lambda\mathbf{p} \qquad \text{for some } 0 \le \lambda \le 1.$$

On line OS, any point from O towards S has the form

$$\vec{OT} = \mu\,\vec{OS} = \mu\left(\mathbf{r} + \frac{1}{2}\mathbf{p}\right) = \mu\mathbf{r} + \frac{\mu}{2}\mathbf{p} \qquad \text{for some } 0 \le \mu \le 1.$$

Since $\mathbf{p}$ and $\mathbf{r}$ are not parallel (they form two sides of a parallelogram), matching coefficients in

$$(1-\lambda)\mathbf{r} + \lambda\mathbf{p} = \mu\mathbf{r} + \frac{\mu}{2}\mathbf{p}$$

gives

$$1 - \lambda = \mu \qquad \text{and} \qquad \lambda = \frac{\mu}{2}.$$

Substituting the second equation into the first,

$$1 - \frac{\mu}{2} = \mu \quad \Longrightarrow \quad 1 = \frac{3\mu}{2} \quad \Longrightarrow \quad \mu = \frac{2}{3}, \quad \lambda = \frac{1}{3}.$$

Therefore

$$\vec{OT} = \frac{2}{3}\mathbf{r} + \frac{1}{3}\mathbf{p}.$$

(ii) From part (i), on line PR we have

$$\vec{OT} = \frac{1}{3}\mathbf{p} + \frac{2}{3}\mathbf{r}.$$

This means T lies one third of the way from R to P, so it divides PR internally in the ratio

$$PT : TR = \frac{2}{3} : \frac{1}{3} = 2 : 1.$$

Equivalently, $\vec{PT} = \vec{OT} - \vec{OP} = \frac{2}{3}\mathbf{r} - \frac{2}{3}\mathbf{p} = \frac{2}{3}(\mathbf{r} - \mathbf{p})$ and $\vec{TR} = \vec{OR} - \vec{OT} = \frac{1}{3}(\mathbf{r} - \mathbf{p})$, so $\vec{PT} = 2\vec{TR}$.

Takeaways 3.32

- Parametric form: Express points on lines using parameters
- Intersection: Equate parametric forms to find intersection point
- Section formula: Use to determine division ratio

3.33 Solution to Problem 33: Two Masses with Pulley

Solution 3.33

The larger mass ($4m$) moves downward with speed v, so the smaller mass ($2m$) moves upward with the same speed. Each mass experiences air resistance of magnitude kv opposing its motion.

(i) Take downward as positive for the $4m$ mass. The forces are weight $4mg$ (down), tension T (up), and resistance kv (up), so Newton's second law gives

$$4mg - T - kv = 4m\,\frac{dv}{dt}.$$

For the $2m$ mass, take upward as positive. The forces are tension T (up), weight $2mg$ (down), and resistance kv (down), so

$$T - 2mg - kv = 2m\,\frac{dv}{dt}.$$

The string is light and the pulley is smooth, so the tension is the same on both masses. Adding the two equations eliminates T:

$$(4mg - T - kv) + (T - 2mg - kv) = 6m\,\frac{dv}{dt},$$

$$2mg - 2kv = 6m\,\frac{dv}{dt}.$$

Dividing by $6m$,

$$\frac{dv}{dt} = \frac{2mg - 2kv}{6m} = \frac{gm - kv}{3m}.$$

(ii) The motion satisfies

$$\frac{dv}{dt} = \frac{gm - kv}{3m}.$$

Separate variables:

$$\frac{dv}{gm - kv} = \frac{dt}{3m}.$$

Integrate both sides:

$$-\frac{1}{k}\ln|gm - kv| = \frac{t}{3m} + C.$$

The masses start from rest, so $v = 0$ when $t = 0$. Substituting gives

$$-\frac{1}{k}\ln(gm) = C.$$

Hence

$$-\frac{1}{k}\ln|gm - kv| = \frac{t}{3m} - \frac{1}{k}\ln(gm).$$

Rearranging,

$$\ln\left|\frac{gm - kv}{gm}\right| = -\frac{kt}{3m}.$$

We are given $v < \dfrac{gm}{k}$, so $gm - kv > 0$ and we may drop the absolute value:

$$\frac{gm - kv}{gm} = e^{-kt/(3m)}.$$

Solving for v,

$$v = \frac{gm}{k}\left(1 - e^{-kt/(3m)}\right).$$

When $t = \dfrac{3m}{k}\ln 2$,

$$v = \frac{gm}{k}\left(1 - e^{-\ln 2}\right) = \frac{gm}{k}\left(1 - \frac{1}{2}\right) = \frac{gm}{2k}.$$

Takeaways 3.33

- Newton's second law: Apply to each mass separately with directions chosen to match the motion
- System of equations: Add the equations to eliminate tension
- Separable ODE: Integrate and use the initial condition $v(0) = 0$ to find velocity as a function of time

3.34 Solution to Problem 34: Integration with Recurrence and Factorial Inequality

Solution 3.34

(i) Let $u = \sin^{2n}(2\theta)$ and $dv = \sin(2\theta)\,d\theta$.
Then $du = 2n\sin^{2n-1}(2\theta)\cdot 2\cos(2\theta)\,d\theta = 4n\sin^{2n-1}(2\theta)\cos(2\theta)\,d\theta$.
And $v = -\frac{1}{2}\cos(2\theta)$.
Using integration by parts:

$$I_n = \left[-\frac{1}{2}\sin^{2n}(2\theta)\cos(2\theta)\right]_0^{\pi/2} + 2n\int_0^{\pi/2}\sin^{2n-1}(2\theta)\cos^2(2\theta)\,d\theta$$

The boundary term is 0. Using $\cos^2(2\theta) = 1 - \sin^2(2\theta)$:

$$I_n = 2n\int_0^{\pi/2}\sin^{2n-1}(2\theta)(1-\sin^2(2\theta))\,d\theta = 2n(I_{n-1} - I_n)$$

Solving: $I_n = 2nI_{n-1} - 2nI_n$, so $(2n+1)I_n = 2nI_{n-1}$.
Therefore: $I_n = \frac{2n}{2n+1}I_{n-1}$.
(ii) $I_0 = \int_0^{\pi/2}\sin(2\theta)\,d\theta = \left[-\frac{1}{2}\cos(2\theta)\right]_0^{\pi/2} = 1$.
Using the recurrence: $I_n = \frac{2n}{2n+1}\cdot\frac{2(n-1)}{2n-1}\cdot\ldots\cdot\frac{2}{3}\cdot 1 = \frac{2^n n!}{(2n+1)\cdot(2n-1)\cdot\ldots\cdot 3\cdot 1}$.
The denominator is $(2n+1)!! = \frac{(2n+1)!}{2^n n!}$.
So: $I_n = \frac{2^n n!\cdot 2^n n!}{(2n+1)!} = \frac{2^{2n}(n!)^2}{(2n+1)!}$.
(iii) Using substitution $x = \sin^2\theta$, so $dx = 2\sin\theta\cos\theta\,d\theta$ and $1 - x = \cos^2\theta$:

$$J_n = \int_0^1 x^n(1-x)^n\,dx = \int_0^{\pi/2}\sin^{2n}\theta\cos^{2n}\theta\cdot 2\sin\theta\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2}\sin^{2n+1}\theta\cos^{2n+1}\theta\,d\theta = 2\int_0^{\pi/2}(\sin\theta\cos\theta)^{2n+1}\,d\theta$$

Using $\sin(2\theta) = 2\sin\theta\cos\theta$: $J_n = 2\int_0^{\pi/2}\left(\frac{1}{2}\sin(2\theta)\right)^{2n+1}\,d\theta = \frac{1}{2^{2n}}I_n$.
So $J_n = \frac{1}{2^{2n}}\cdot\frac{2^{2n}(n!)^2}{(2n+1)!} = \frac{(n!)^2}{(2n+1)!}$.
(iv) Since $0 \le x^n(1-x)^n \le 1$ for $x \in [0,1]$ (maximum occurs at $x = 1/2$), we have $J_n \le \int_0^1 1\,dx = 1$.
Therefore: $\frac{(n!)^2}{(2n+1)!} \le 1$, so $(n!)^2 \le (2n+1)!$.
Multiplying both sides by 2^{2n}: $(2^n n!)^2 = 2^{2n}(n!)^2 \le 2^{2n}(2n+1)!$.
Actually, we need $(2^n n!)^2 \le (2n+1)!$.
From $J_n = \frac{(n!)^2}{(2n+1)!} \le 1$, we get $(n!)^2 \le (2n+1)!$.
But we need $(2^n n!)^2 = 2^{2n}(n!)^2 \le (2n+1)!$.
This requires $2^{2n} \le \frac{(2n+1)!}{(n!)^2}$, which is true for $n \ge 1$ (can be verified).
Actually, a direct approach: $(2n+1)! = (2n+1)(2n)(2n-1)\ldots(3)(2)(1)$.
We have $(2^n n!)^2 = 2^{2n}(n!)^2 = 2^{2n}(1\cdot 2\cdot\ldots\cdot n)^2$.
Comparing term by term, $(2n+1)!$ contains factors that are at least as large, so the inequality holds.

Takeaways 3.34

- Integration by parts: Use to derive recurrence relations
- Double factorial: $(2n+1)!! = \frac{(2n+1)!}{2^n n!}$
- Substitution: $x = \sin^2\theta$ relates polynomial and trigonometric integrals
- Bounding integrals: Use maximum value of integrand to bound the integral

3.35 Solution to Problem 35.1 (parts a–d)

Solution 3.35

Throughout, $A_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x\,dx$ and $B_n = \int_0^{\frac{\pi}{2}} x^2 \cos^{2n} x\,dx$.

(a) Write $\cos^{2n} x = \cos^{2n-1} x \cos x$ and integrate by parts with $u = \cos^{2n-1} x$, $dv = \cos x\,dx$:

$$A_n = \left[\cos^{2n-1} x \sin x\right]_0^{\frac{\pi}{2}} + (2n-1)\int_0^{\frac{\pi}{2}} \sin^2 x \cos^{2n-2} x\,dx.$$

The boundary term is zero. Using $\sin^2 x = 1 - \cos^2 x$,

$$A_n = (2n-1)(A_{n-1} - A_n) \implies nA_n = \frac{2n-1}{2}A_{n-1}.$$

(b) Integrate A_n by parts with $u = \cos^{2n} x$, $dv = dx$:

$$A_n = \left[x\cos^{2n} x\right]_0^{\frac{\pi}{2}} + 2n\int_0^{\frac{\pi}{2}} x \sin x \cos^{2n-1} x\,dx = 2n\int_0^{\frac{\pi}{2}} x\sin x\cos^{2n-1} x\,dx.$$

(c) From part (b), divide by n^2 and integrate $\int_0^{\frac{\pi}{2}} x\sin x\cos^{2n-1} x\,dx$ by parts with $u = \sin x$, $dv = x\cos^{2n-1} x\,dx$. The boundary terms vanish, giving

$$\frac{A_n}{n^2} = \frac{2n-1}{n}B_{n-1} - 2B_n.$$

(d) From part (c), divide by A_n and multiply by 2:

$$\frac{2}{n^2} = \frac{2(2n-1)}{n}\frac{B_{n-1}}{A_n} - 4\frac{B_n}{A_n}.$$

Part (a) gives $A_{n-1} = \frac{2nA_n}{2n-1}$, so $\frac{2(2n-1)}{n}\frac{B_{n-1}}{A_n} = 4\frac{B_{n-1}}{A_{n-1}}$. Hence

$$\frac{1}{n^2} = 2\left(\frac{B_{n-1}}{A_{n-1}} - \frac{B_n}{A_n}\right).$$

Takeaways 3.35

- Integration by parts on $\cos^{2n} x$ yields the reduction formula in part (a)
- Parts (b) and (c) link A_n to the auxiliary integral B_n
- Part (d) rewrites $1/n^2$ as a telescoping difference of ratios B_k/A_k

3.36 Solution to Problem 35.2 (parts e–g)

Solution 3.36

Recall $A_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x\,dx$ and $B_n = \int_0^{\frac{\pi}{2}} x^2 \cos^{2n} x\,dx$.

(e) Summing part (d) from $k = 1$ to n telescopes:

$$\sum_{k=1}^{n} \frac{1}{k^2} = 2\left(\frac{B_0}{A_0} - \frac{B_n}{A_n}\right).$$

Now $A_0 = \frac{\pi}{2}$, $B_0 = \frac{\pi^3}{24}$, so $\frac{B_0}{A_0} = \frac{\pi^2}{12}$. Hence

$$\sum_{k=1}^{n} \frac{1}{k^2} = \frac{\pi^2}{6} - 2\frac{B_n}{A_n}.$$

(f) On $0 \le x \le \frac{\pi}{2}$, $\sin x \ge \frac{2}{\pi}x$, so

$$\cos x = \sqrt{1 - \sin^2 x} \le \sqrt{1 - \frac{4x^2}{\pi^2}} \quad \Longrightarrow \quad \cos^{2n} x \le \left(1 - \frac{4x^2}{\pi^2}\right)^n.$$

Therefore

$$B_n \le \int_0^{\frac{\pi}{2}} x^2 \left(1 - \frac{4x^2}{\pi^2}\right)^n dx.$$

(g) Integrate by parts on the right-hand side of (f) with $u = x$, $dv = x\left(1 - \frac{4x^2}{\pi^2}\right)^n dx$. Since $\int x\left(1 - \frac{4x^2}{\pi^2}\right)^n dx = -\frac{\pi^2}{8(n+1)}\left(1 - \frac{4x^2}{\pi^2}\right)^{n+1}$, the boundary term vanishes and

$$\int_0^{\frac{\pi}{2}} x^2 \left(1 - \frac{4x^2}{\pi^2}\right)^n dx = \frac{\pi^2}{8(n+1)} \int_0^{\frac{\pi}{2}} \left(1 - \frac{4x^2}{\pi^2}\right)^{n+1} dx.$$

Takeaways 3.36

- Telescoping part (d) expresses the partial sum in terms of B_n/A_n
- The inequality $\sin x \ge 2x/\pi$ bounds $\cos^{2n} x$ from above
- Part (g) evaluates the bounding integral by a further integration by parts

3.37 Solution to Problem 35.3 (parts h–j)

Solution 3.37

Recall $A_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x\,dx$ and $B_n = \int_0^{\frac{\pi}{2}} x^2 \cos^{2n} x\,dx$.

(h) Combining parts (f) and (g),

$$B_n \le \frac{\pi^2}{8(n+1)} \int_0^{\frac{\pi}{2}} \left(1 - \frac{4x^2}{\pi^2}\right)^{n+1} dx.$$

Substitute $x = \frac{\pi}{2}\sin t$, so $dx = \frac{\pi}{2}\cos t\,dt$ and $1 - \frac{4x^2}{\pi^2} = \cos^2 t$:

$$\int_0^{\frac{\pi}{2}} \left(1 - \frac{4x^2}{\pi^2}\right)^{n+1} dx = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \cos^{2n+3} t\,dt.$$

Hence

$$B_n \le \frac{\pi^3}{16(n+1)} \int_0^{\frac{\pi}{2}} \cos^{2n+3} t\,dt \le \frac{\pi^3}{16(n+1)} A_n,$$

since $\cos^{2n+3} t \le \cos^{2n} t$ on $\left[0, \frac{\pi}{2}\right]$.

(i) From part (e),

$$\sum_{k=1}^{n} \frac{1}{k^2} = \frac{\pi^2}{6} - 2\frac{B_n}{A_n}.$$

Part (h) gives $\frac{B_n}{A_n} \le \frac{\pi^3}{16(n+1)}$ and $B_n > 0$, so

$$\frac{\pi^2}{6} - \frac{\pi^3}{8(n+1)} \le \sum_{k=1}^{n} \frac{1}{k^2} < \frac{\pi^2}{6}.$$

(j) As $n \to \infty$, the error term $\frac{\pi^3}{8(n+1)} \to 0$. By the squeeze theorem,

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

Takeaways 3.37

- The substitution $x = \frac{\pi}{2}\sin t$ converts the bound to a Wallis-type integral
- Part (i) sandwiches the partial sum between two expressions differing by $O(1/n)$
- The squeeze theorem in part (j) gives Euler's value $\pi^2/6$ for the Basel sum

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